MIL-C-13962B

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MILITARY SPECIFICATION

CRUSHING, SCREENING, AND WASHING PLANT, DIESEL-ENGINE

AND ELECTRIC-MOTOR DRIVEN, 225-TONS PER HOUR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a combined crushing, screening, and washing plant consisting of mobile crushing units and auxiliary mobile equipment on pneumatic rubber-tired road wheels, compatible for towing behind various trucks and truck tractors.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

W-C-586	- Conduit Outlets and Entrance Caps, Electrical; Cast Metal - for Shore Use.
FF-B-171	- Bearings, Ball, Annular, General Purpose.
FF-B-185	- Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187	- Bearing, Roller, Tapered.
QQ-S-781	- Steel Strapping Flat.
QQ-S-790	- Steel Strapping, Round (Bare and Zinc-Coated).
SS-R-406	- Road and Paving Materials; Methods of Sampling and Testing.

FSC 3820

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TT-P-664	- Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
TT-V-121	- Varnish, Spar, Water-Resisting.
UU-T-81	- Tags, Shipping and Stock.
WW-C-566	- Conduit, Metal, Flexible.
WW-C-581	- Conduit, Metal, Rigid; and Coupling, Elbow, and Nipple, Electrical Conduit: Zinc-Coated.
ZZ-B-206	- Belting; Conveyor (Rubber and Synthetic Rubber.)
ZZ-H-561	- Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water.
KKK-L-370	- Lining, Friction, (Clutch and Brake; Metallic, Metal-Ceramic and Semimetallic).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-C-650	- Crates, Wood, Open and Covered.
PPP-T-60	- Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.

Military

MIL-V-3	- Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.
MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture-and Fungus-Resistant, (for the Treatment of Communications, Electronic and Associated Electrical Equipment).
MIL-P-514	- Plates, Identification; Transportation-Data and Blank.
MIL-I-631	- Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-H-775	- Hose, Rubber or Fabric (Including Tubing); and Fittings, Nozzles and Strainers, Packaging of.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-C-3432	- Cable and Wire, Electrical (Power and Control; Semi-Flexible, Flexible and Extra Flexible, 300 and 600 Volts).
MIL-F-3541	- Fittings, Lubrication.
MIL-C-3774	- Crates, Wood; Open 12,000- and 16,000-Pound Capacity.
MIL-G-3859	- Grease Guns, Hand, High-Pressure, Lever-Operated (15-Ounce and 21-Ounce Capacity).

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MIL-H-3992 - Hose and Hose Assembly, Rubber; Air and Vacuum Brake, Automotive.
MIL-S-8660 - Silicone Compound.
MIL-E-10062 - Engines: Preparation for Shipment and Storage of.
MIL-P-10360 - Pumps, Centrifugal, For Water.
MIL-G-10924 - Grease, Automotive and Artillery.
MIL-E-11276 - Engines, Diesel: Industrial, Medium- and High-Speed, General Specification for.
MIL-S-12514 - Starters, Motor; Across-the-Line and Reduced Voltage Types, A-C Induction, 3/4 to 100-HP, Manual and Electric.
MIL-C-13486 - Cable, Special Purpose, Electrical: Low-Tension, Heavy Duty Single-Conductor and Multiconductor.
MIL-P-13983 - Paint, Temporary, Lusterless, Gasoline Removable.
MIL-I-15126 - Insulation Tape, Electrical, Pressure Sensitive Adhesive and Pressure Sensitive Thermosetting Adhesive.
MIL-H-15424 - Hand Tools; Packaging of.
MIL-E-16298 - Electric Machines having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-V-16721 - Valves, Foot, Low Pressure (for Water).
MIL-L-21260 - Lubricating Oil, Internal-Combustion Engine, Preservative.
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-C-52058 - Chains, Roller, Power Transmission, and Conveyor.
MIL-A-52363 - Air Cleaners, Intake: Dry-Type (for Internal Combustion Engine).
MIL-G-52461 - Generator Set, Diesel-Engine-Driven, 100-KW, Alternating-Current, Liquid-Cooled, Portable, Skid-Mounted, Multipurpose.
MIL-E-55301 - Electromagnetic Compatibility.

STANDARDS

Federal

FED. STD. No. 595 - Colors

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.

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MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-619	- Unified Soil Classification System for Roads, Airfields, Embankments and Foundations.
MIL-STD-642	- Identification Marking of Administrative, Combat, and Tactical Transport Vehicles.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1400	- Engines, Gasoline or Diesel, Methods of Test.
MS16961	- Troughing Carrier Assembly, 4 Inch Diameter Roll, 4-5/8 Inch Backing; 24 and 30 Inch Belt Width.
MS16962	- Troughing Carrier Assembly, 4 Inch Diameter Roll, 6-3/4 Inch Backing; 24 and 30 Inch Belt Width.
MS16963	- Troughing Carrier Assembly, Impact, 4 Inch Diameter Rubber Roll, 4-5/8 Inch Backing, 24 and 30 Inch Belt Width.
MS16964	- Return Roll Assembly, 4 Inch Diameter Roll, For 4-5/8 Inch Backing Trough Carrier; 24 and 30 Inch Belt Width.
MS16965	- Return Roll Assembly, 4 Inch Diameter Roll, For 6-3/4 Inch Backing Trough Carrier; 24 and 30 Inch Belt Width.
MS21322	- Clamps, Loop-Aluminum-Plain and Cushioned.
MS21333	- Clamps, Loop-Steel-Plain and Cushioned.
MS21334	- Clamps, Loop-Double, Cable or Tube, Steel, Plain and Cushioned.
MS24207	- Coupling, Grease Gun.
MS27142	- Connector, Plug, Electrical-Pin Contact No. 12, 14, and 16 A.W.G., Waterproof.
MS27143	- Connector, Plug, Electrical-Socket Contact, No. 12 A.W.G., Waterproof.
MS27144	- Connector, Plug, Electrical-Socket Contact, No. 14 and 16 A.W.G., Waterproof.
MS27145	- Connector, Plug, Electrical- No. 14 A.W.G., Double Contact, Waterproof.
MS27146	- Connector, Receptacle, Electrical- No. 12, 14, and 16 A.W.G., Double Contact, Waterproof.
MS27147	- Adapter, Connector - 3 Way, No. 12, 14 and 16 A.W.G., Waterproof.
MS27148	- Pin Contact, Electrical Connector- No. 12, 14, and 16 A.W.G.
MS33800	- Connector, Receptacle, Electrical-Pin Contact, No. 12, 14 and 16 A.W.G., Waterproof.

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MS35387 - Reflector, Indicating, Clearance.
MS35388 - Tire, Pneumatic, Passenger Car, Light Truck, Truck and Bus.
MS35392 - Inner Tube, Pneumatic Tire.
MS35423 - Light, Marker, Clearance-Service.
MS35424 - Light, Marker, Clearance-Blackout, 24 Volt.
MS35643 - Gasket: Cap, Fuel Tank.
MS35644 - Filler Neck Assembly, Fuel Tank - Military Vehicles.
MS35645 - Cap, Fuel Tank, Tactical Vehicle, Fording.
MS35746 - Coupling, Automotive-Air Brake Lines.
MS35773 - Radiators, Engine Cooling, Industrial.
MS35844 - Plug, Machine Thread; Magnetic (Drain).
MS35865 - Axle, Vehicular, Non-Driving, Single, for Tactical Wheeled Vehicles.
MS39020 - Band, Marker, Blank.
MS39062 - Circuit Breakers: Thermal, 30 Volt DC, Automatic, Waterproof, 15 to 30 Amp.
MS39127 - Tray Screen Sizes and Edging for Aggregate Screens.
MS51118 - Pintle Assembly, Touring - 40,000 Lbs. Capacity, Manual Release.
MS51330 - Stop Light-Taillight, Vehicular-24 Volt, Blackout Tail, Blackout Stop.
MS51331 - Guard, Splash, Wheel.
MS51336 - Lunette-Coupler, Drawbar, Ring.
MS51337 - Coupler, Drawbar, Ring-Heavy Duty.
MS51339 - Coupler, Drawbar, Ring-Offset.
MS53004 - Valve, Relay, Emergency-Breakaway, Air Brake.
MS53007 - Plate, Identification-Emergency and Service.
MS53036 - Kingpin, Fifth Wheel, 2 Inch.
MS53038 - Kingpin, Fifth Wheel - Universal.
MS53039 - Socket, Kingpin, Fifth Wheel - Universal.
MS53040 - Retainer, Kingpin, Fifth Wheel - Universal.
MS53044 - Wheel, Pneumatic Tire - Disk Type, with Ring, for Tactical Wheeled Vehicles.
MS53063 - Indicator, Air Cleaner, Intake; Restriction Mechanical Type.
MS53091 - Landing Gear, Retractable-Semitrailer, Folding Leg, 50,000 Lb. Static Capacity.
MS53093 - Landing Gear, Retractable - Semitrailer, Low Bed, 60,000 Lb. Static Capacity.

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MS75021	- Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 24 Volt, Waterproof.
MS500008	- Truck, Tractor, 5 Ton, 6 X 6, M52.
MS500010	- Truck, Dump, 5 Ton, 6 X 6, M51.

DRAWINGS

MO(ATAC)

8347201	- Schematic Wiring Diagram, Semitype Trailer.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN WELDING SOCIETY

Standard Qualification Procedure.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N. Y. 10017.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

AB1 - Circuit Breakers, Molded Case.
MG1 - Standards Publication for Motors and Generators.
IC1 - Standards for Industrial Control.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, N. Y. 10017.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

A47 - Malleable Iron Castings.
A128 - Austenitic Manganese-Steel Castings.
A148 - High-Strength Steel Castings for Structural Purposes.
A193 - Alloy-Steel Bolting Materials for High-Temperature Service.
A536 - Ductile Iron Castings.
C170 - Compressive Strength of Natural Building Stone.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

STATE OF CALIFORNIA DEPARTMENT OF MOTOR VEHICLES

California Vehicle Code.

(Application for copies should be addressed to the State of California Department of Motor Vehicles, P.O. Box 1319, Sacramento 6, California.)

AMERICAN STANDARDS ASSOCIATION

B6.6 - Gear Tolerances and Inspection.
B55.1 - Specifications for Multiple V-Belt Drives.

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N. Y. 10016.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 485 Lexington Avenue, New York, N. Y. 10017.)

RUBBER MANUFACTURERS ASSOCIATION, INC.

Standard Specifications for Drives Using Narrow V-Belts (3V, 5V, and 8V Cross Sections).

(Application for copies should be addressed to the Rubber Manufacturers Association, Inc., 444 Madison Avenue, New York, N. Y. 10022.)

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FRICTION MATERIALS STANDARDS INSTITUTE, INC.

Automotive Data Book.

(Application for copies should be addressed to Friction Materials Standards Institute, Inc., 370 Lexington Avenue, New York, N. Y. 10017.)

TIRE AND RIM ASSOCIATION, INC.

Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., Comand Building, 34 North Hawkins Avenue, Akron, Ohio 44313.)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N. Y. 10016.)

AMERICAN TRUCKING ASSOCIATION

E-3 - Recommended Equipment Specification.
National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, 1616 P Street NW, Washington, D. C. 20036.)

INTERSTATE COMMERCE COMMISSION

Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington D. C. 20402.)

3. REQUIREMENTS

3.1 Description. The combined crushing, screening, and washing plant shall consist of mobile crushing units and auxiliary mobile equipment on pneumatic rubber-tired road wheels, compatible for towing behind various trucks and tractors as specified herein, at speeds up to 30 miles per hour.

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Each crushing unit shall be furnished complete with the necessary power units, electric motors, and conveyors for receiving the raw materials and delivering the finished product to the succeeding unit or to stock pile. All necessary detail parts shall be included to make the combined plant complete for the performance of the functions specified herein.

3.1.1 Units. The combined crushing, screening, and washing plant shall produce crushed rock and crushed sand from rock quarries or gravel pits. The combined plant shall consist of the following units:

Unit No. 1. Crusher, jaw-type: diesel-driven; not less than 30-inch by 40-inch opening, 200 to 400 tons per hour (tph), for primary crushing of quarry material. Semitrailer type with dolly converter (see 3.25.3).

Unit No. 2. Crusher, roll-type, secondary: diesel driven, with motor drives on screen and conveyors, 440-volt, 60-cycle, 3-phase: not less than 54-inch diameter by 24-inch face rolls, 200 to 400 tph, for secondary crushing of quarry material. Semitrailer type with dolly converter (see 3.25.4).

Unit No. 3. Conveyor, belt, transfer: motor driven, 440-volt, 60-cycle, 3-phase; 30-inch by 50-foot, with hydraulic cradle; for transfer of crushed rock and fines from unit No. 2 or unit No. 7 to unit No. 4. Cart trailer type (see 3.25.5).

Unit No. 4. Crusher, screen and washer, roll-type: diesel driven, with motor drive on conveyor, 440-volt, 60-cycle, 3-phase; not less than 40-inch diameter by 22-inch face rolls, 40 to 250 tph, for crushing, screening and washing of material. Two units may be substituted for this unit provided the same crushing, screening, and washing capacity is available, and no change is required in the flow of material. Semitrailer type with dolly converter (see 3.25.6).

Unit No. 5. Control panel for distributing the power of unit No. 15 to the electric motors (see 3.25.7).

Unit No. 6. Crusher, roll-type, intermediate: diesel driven, with motor drive on screen, 440-volt, 60-cycle, 3-phase, not less than 40-inch diameter by 22-inch face rolls, 40 to 250 tph for crushing of oversize material in quarry setup to obtain additional fines if required. Semitrailer type with dolly converter (see 3.25.8).

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Unit No. 7. Conveyor, belt: motor driven, 440-volt, 60-cycle, 3-phase, with hydraulic cradle, 30-inch by 35-foot, for return of material from unit No. 6 to unit No. 3 and unit No. 4. Cart trailer type (see 3.25.9).

Unit No. 8. Dehydrator, sand, twin-screw: motor driven, 440-volt, 60-cycle, 3-phase, 22-inch diameter by 20-foot long, 40 to 90 tph, for dewatering of sand from unit No. 4. Semitrailer type with dolly converter or full trailer (see 3.25.10).

Unit No. 9. Feeder, reciprocating-plate: chain-driven, 30-inch wide, with 30-inch by 21-foot electric-motor-powered belt conveyor, for feeding material to unit No. 2 or unit No. 3 in a gravel-pit setup. Only the conveyor is used to feed material from the unit No. 4 to unit No. 10 in a quarry setup (see 3.25.11).

Unit No. 10. Scrubber and washer, gravel, twin-paddle-type: diesel driven; 80 to 120 ton, with 7- by 20-foot tank, for additional washing of coarse aggregate from unit No. 4. Semitrailer type with dolly converter (see 3.25.12).

Unit No. 11. Pump unit, centrifugal: gasoline-engine-driven, trailer-mounted on pneumatic rubber-tired wheels. Two units per plant to supply water to units No. 4, No. 8, and No. 10. When the second pump is used, it will be in series (see 3.25.13).

Unit No. 12. Conveyor, belt, coarse material: motor-driven, 440-volt, 60-cycle, 3-phase, 30-inch by 50-foot, with hydraulic cradle, for transfer of crushed coarse material from unit No. 4 or unit No. 10 to stock pile. Cart trailer type (see 3.25.14).

Unit No. 13. Conveyor, belt, sand: motor-driven, 440-volt, 60-cycle, 3-phase, 30-inch by 50-foot, with hydraulic cradle, for transfer of sand from unit No. 8 to stock pile. Cart trailer type (see 3.25.15).

Unit No. 14. Piping equipment: consisting of equipment for washer, dehydrator, and pumps, including 6-inch by 30-foot suction hose, ponton, waste flumes, and all other accessories necessary to complete the water-supply system for the entire plant (see 3.25.16).

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Unit No. 15. Electric generator set: diesel driven, portable, skid mounted, 100-kw, for the source of electric power for various electric motors in the plant (see 3.25.17).

3.1.2 Weights and dimensions. The gross vehicle weight (GVW) of each unit shall include the weight of the trailer or semitrailer with all machinery, equipment, accessories, attachments, and lubricants. Component and vehicular ratings shall not be raised to conform to the requirements of this specification. The physical characteristics of each unit prepared for transport, shall be as specified in table I, with the unit on level ground and with the inflation pressure of the tires as recommended in the Tire and Rim Association Yearbook for the imposed GVW loading.

Table I. Weights and Dimensions

Characteristic	Requirement
Overall length	56 feet, maximum
Overall width	120 inches, maximum
Overall height	126 inches, maximum
Ground clearance, exclusive of suspension	12 inches, minimum
Semitrailer type:	
GVW	100,000 pounds, max.
Fifth-wheel load	30,000 pounds, max.
Axle load, each	18,000 pounds, max.
Upper fifth-wheel height, range	54 to 60 inches
Kingpin location, range (SAE J701)	24 to 36 inches
Swing radius clearance (SAE J701)	60 inches, maximum
Landing wheel clearance (SAE J701)	88 inches, minimum
Cart-trailer type:	
GVW	14,000 pounds, max.
Vertical lunette load, range	10 to 15 percent of GVW
Axle load, each	10,000 pounds, max.
Tow bar eye height, range (SAE J849)	31 to 37 inches

3.2 Preproduction model. The supplier shall furnish a plant within the time frame specified in the contract or purchase order to prove, prior to starting production, that his production methods and choice of design

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detail will produce plants that comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the preproduction model during production shall be subject to the approval of the contracting officer. Approval of the preproduction model by the activity concerned shall not relieve the supplier of his obligation to furnish plants conforming to this specification.

3.3 Initial production. Unless otherwise specified (see 6.2), the supplier shall furnish to the Government a plant for testing as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4.1 Stress limitations. The plant shall be constructed so that the maximum stress imposed on any structural component shall not exceed 60 percent of the yield point.

3.4.2 Bearings. Unless otherwise specified herein, all bearings, except power-unit bearings, shall be of antifriction type. All bearing housings, except in the power units and jaw crushers, shall be equipped with pressure-lubricating fittings. Ball and roller bearings, except power unit bearings, shall conform to FF-B-171, FF-B-185, and FF-B-187, as applicable, and shall be provided with labyrinth seals. Conveyor load supporting and return idler roller bearings may be of the sealed-for-life type and shall otherwise conform to FF-B-171.

3.4.3 Wire screen. All wire screen shall be 4-foot wide screen trays conforming to MS39127; except that the length of the trays used on the top, second, and fourth decks on unit No. 2 shall be 72 inches.

3.4.4 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. Maximum practicable use shall be made of interchangeable hardware and fastening devices.

3.5 Construction. The construction of the plant shall permit the units to be arranged for either a gravel pit setup or a rock-quarry setup for crushing large rock for producing additional fines, for returning the fines for screening and washing, and for washing out shale and clay. The construction of the separate units of the plant shall permit arrangement of the units in the different combinations to meet requirements for the finished product when set up in a quarry or a gravel pit.

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3.5.1 Safety. The safety characteristics of all units shall reduce hazards to personnel and property to a minimum.

3.5.1.1 Guards. All drives and moving parts that are 6 feet 6 inches or less from the ground shall be completely guarded or protected to prevent injury to the operating personnel. All drives and moving parts above 6 feet 6 inches which could be injurious to the operator shall be completely guarded or protected with the exception of such parts as jaws on the primary crusher and the moving conveyor belts which in commercial use are left unguarded. "Completely guarded" shall mean guarded from sides, front, top, and bottom as required. Guards shall be provided to prevent damage to all V-belt and chain drives from falling material.

3.5.1.2 Set screws and collars. All points using set screws shall use allen set screws with the heads below the surface when in position for operation. All collars shall be smooth and of a shape that will not be hazardous to operating personnel.

3.5.1.3 Operator's platforms. A steel platform made of expanded metal shall be provided where an operator is required. The steel platform shall be equipped with handrails and shall be of sufficient size to permit the operator to manipulate the controls.

3.5.1.3.1 Controls. All controls shall be mounted so that they are accessible to the operator in his operating position. Operating personnel shall vary from the small man, clothed, through the large man, arctic clothed, in accordance with SAE J833.

3.5.1.4 ICC regulations. Each unit shall conform to all large trailer requirements of the ICC Motor Carrier Safety Regulations, except as specified herein.

3.6 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance and repair utilizing common tools and special tools associated with crushing, screening, and washing plants. The replacement and adjustment of components, parts, and accessories shall be accomplished with minimum drainage requirements and with minimum disturbance to other components and parts of the crushing, screening, and washing plant.

3.7 Transportability.

3.7.1 Performance characteristics. Each unit shall conform to the transport performance characteristics specified herein when prepared for transport and towed by a prime mover.

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3.7.1.1 Highway operation. Each unit shall travel over smooth, hard-surfaced primary highways at speeds up to 30 mph.

3.7.1.2 Cross-country operation. Each unit shall withstand towing by truck tractor over unimproved roads and open, rolling, and hilly cross-country terrain at speeds up to 10 mph. Each unit shall travel on grades up to 60 percent; mud, sand, and snow; and transport incident to intended operation sites, when towed by a crawler tractor. Each unit shall ford hard-bottomed bodies of water up to 30 inches in depth for periods up to 15 minutes.

3.7.1.3 Trailing and turning ability. Each unit shall trail the towing vehicle without weaving. Each unit shall follow the towing vehicle without exceeding 75 percent of the ICC towed-vehicle tolerance. Jackknife turns up to an angle of 90 degrees for semitrailers and turns up to 60 degrees for dolly-semitrailer combinations as measured from straight-ahead to the right and to the left shall not cause damage to or interference between the towed unit, dolly, and the towing vehicle.

3.7.1.4 Braking ability. For each unit, the combined service brakes of the towing and towed vehicle shall control, decelerate, and stop the combination in accordance with ICC requirements, except that the stopping distance shall be accurately measured from initial movement of the brake pedal to stopping point. When actuated, the emergency breakaway system shall produce smooth emergency brake application.

3.7.1.5 Parking ability. When the emergency-parking brakes are applied, the wheels shall remain locked on grades up to 40 percent despite the depletion of the compressed air supply.

3.7.2 Rail. The plant shall withstand impact forces encountered in rail transport without breakage or permanent deformation. Sectionalization for rail transportation shall be within the dimensional limits shown on figure 1. Sectionalization and reassembly shall each be accomplished in not more than 6 hours without cutting or welding.

3.7.2.1 Lifting and tiedown attachments. All sections of the plant which must be removed to meet the requirements of 3.7.2 shall be provided with lifting and tiedown attachments conforming to 3.16 and 3.17.

3.8 Power units.

3.8.1 Engine. Unless otherwise specified (see 6.2), the engines for units 1, 2, 4, 6, and 10 shall be of the naturally aspirated type. Each engine, including all systems, components, accessories and auxiliaries, shall conform to MIL-E-11276, except as specified herein.

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3.8.1.1 Power and speed rating. The power and speed rating of the engine shall be such that operation of the plant under any of the operational requirements and conditions specified herein will not require horsepower in excess of the continuous horsepower rating of the engine for the applicable governed speed, as established by the Qualified Products List, QPL-11276. The maximum engine torque at rated governed speed shall not exceed 90 percent of the peak engine torque available at a lower speed as determined by test method series 2000 of MIL-STD-1400 (see 4.6.2.2).

3.8.1.2 Fuel tank. A fuel tank in accordance with MIL-E-11276 shall be furnished. The tank shall be furnished with a gasket, filler neck, and filler neck cap in accordance with MS35643, MS35644, and MS35645, respectively. The tank shall have a minimum capacity sufficient for not less than 10 hours of operation in any operating condition specified herein. Fuel capacity shall be based on 85 percent of the fuel consumption rate obtained during conduct of tests in test method series 2000 of MIL-STD-1400, at the applicable horsepower and governed speed.

3.8.1.3 Battery charging system. A 40-ampere battery charging system shall be furnished.

3.8.1.4 Instrumentation. An instrument panel, with all instruments as specified in MIL-E-11276 (except tachometer), shall be furnished.

3.8.1.5 Control systems. All control systems as specified in MIL-E-11276 shall be furnished. All controls shall be of the warning-light type, except the overspeed control which shall be of the engine-shut-off type.

3.8.1.6 Cooling system. A complete cooling system in accordance with MIL-E-11276 shall be furnished. The radiator shall conform to MS35773.

3.8.1.7 Air cleaner. The air cleaner shall be of the dry type conforming to MIL-A-52363, class III. A rain cap shall be furnished. The air cleaner shall be provided with a restriction indicator conforming to MS53063-3.

3.8.1.8 Engine housing. Each engine shall be provided with a housing in accordance with MIL-E-11276. Weathertight seals shall be provided for operating doors and panels of the housing. Material for seals shall conform to ZZ-R-765, class and grade optional, and shall be covered with nylon or other suitable material with a securing lip. Seals shall be riveted to the doors, panels, or housing after reinforcing with backup strip to prevent the rivet heads from pulling through the securing lip.

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3.8.1.9 Battery box. A battery box shall be furnished in accordance with MIL-E-11276.

3.8.1.10 Clutch and clutch control. Each engine shall be equipped with a clutch. The clutch shall be accessible for lubrication, adjustment, and replacement of parts. The clutch lining shall conform to KKK-L-370, type II, and shall be installed in accordance with the lining manufacturer's instructions. Engagement or disengagement of the clutch at any engine speed up to maximum governed speed shall not cause damage to the clutch. The clutch capacity shall be not less than 150 percent of the maximum torque available from the engine as determined by test method series 2000 of MIL-STD-1400. The clutch control shall be located so as to be accessible from the operator's platform and also be accessible from the ground.

3.8.1.11 Cranking system. An electric cranking system in accordance with MIL-E-11276 shall be furnished.

3.8.2 Electric motors. The electric motors shall be in accordance with NEMA Standard MG1, rated at 1,800 revolutions per minute (rpm), 440-volt, three-phase, 60-cycle, ball bearing, squirrel-cage-induction class, continuous rating, totally enclosed, fan cooled, design "C", integral horsepower, class "A" insulation, temperature rise 55° C. (131° F.). Bearings shall be in accordance with FF-B-171 and shall be a double-seal type packed with grease in accordance with MIL-G-23827.

3.8.2.1 Electric motor controls. Each motor shall be provided with a motor starter conforming to MIL-S-12514, type I, class 2, style B, size as applicable and shall be 3-pole, alternating current, magnetic nonreversing across-the-line, and rated for 440-volt, 60-cycle operation. The enclosures shall be watertight type IV and dusttight type V in accordance with NEMA Standard ICl and shall be provided with start-stop pushbuttons in the covers for the forward direction of rotation only. Provision shall be made for opening the main contactor (see 3.25.7) from master OFF pushbutton stations located at each motor-starting station, and from a vantage point on both the primary jaw-crusher (unit No. 1) and the first roll crusher (unit No. 2). The master OFF control switch may be incorporated in the corresponding motor starter or may be furnished in a separate weatherproof enclosure. The overload protection shall be of the hand-reset type. Fuses will not be acceptable as overload devices. Each motor starter with its master OFF control shall be mounted as an integral part of its unit and shall be installed so that it will not be damaged by vibration when the unit on which it is mounted is operating. Cables shall be permanently connected to the motor starter and master OFF control through a watertight packing gland. Provision shall be made for mounting the required power and control cables on each unit adjacent to the motor starter.

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3.8.2.2 Power cable, reels, and ground rods. The main feeder cable shall consist of two identical 100-foot lengths conforming to MIL-C-3432, type CO-03HLF (3/00-3/5R) 1750. Each length shall have a 4-pole male connector on one end and a 4-pole female connector on the other end. The cables shall have safety chains and strain collars. The safety chain shall be attached to the male end of each cable. Each connector shall have a protective cap attached with sash chain. Each cable assembly shall be equal to and interchangeable with Joy Manufacturing Company assembly No. X8617-25. Two manually operated reels each containing a 100-foot length of the main feeder cable shall be permanently mounted on the plant. One reel shall be mounted on unit No. 1 and the other reel on unit No. 2. The reels shall be of steel construction, ratchet type, and provision shall be made to secure the cables on the reels. Three ground-rod assemblies for grounding the main power source shall be provided. Each assembly shall consist of a driving bolt and three 3-foot sections of 5/8-inch-diameter steel rod, each section with a coupling. The rods shall consist of a steel core with a copper surface bonded to the core by fusion or electrolytic means. Each end of each rod section shall be threaded, and one end of one rod section shall be provided with a cone point. Figure 2 shows the cone-pointed rod section, one coupling, and the driving bolt. Couplings shall be of copper alloy. Driving bolts shall be steel with a square end. All threads shall be 5/8 inch-11NC. A ground clamp shall be provided with each assembly to accommodate the connection of a flexible AWG No. 6 ground wire. The ground wire shall be 15 feet long. The ground-rod assemblies, clamps, and wires shall be carried in a container permanently fastened to the applicable unit and marked to identify its contents.

3.8.2.2.1 Power cable adaptor assembly. A power cable adaptor assembly shall be provided. The assembly shall consist of a 4-pole female connector on one end of a 5-foot-length of cable conforming to MIL-C-3432, type CO-03HLF (3/00-3/5R) 1750. A strain collar shall be provided to accomodate the safety chain on the male end of the main feeder cable. The adaptor shall have a protective cap attached with sash chain. The other end of the feeder cable shall have 12 inches of the cable jacket removed and shall have 1 inch of insulation stripped from each conductor. The assembly shall be equal to and interchangeable with Joy Manufacturing Company assembly No. X8617-24.

3.8.2.3 Motor supply cable assemblies. Motor supply cables shall conform to MIL-C-3432, type CO-04HLF (4/10) 0767, and shall be provided with a male connector equal to and interchangeable with Joy Manufacturing Co. part No. 441-M for mounting with the corresponding receptacle (see 3.25.7.3).

3.8.2.4 Control cable assemblies. Control cables shall be 2-conductor or 3-conductor as required conforming to MIL-C-3432, class HL, flexibility designation F, size of conductors to carry 125 percent of operating current.

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Cables shall be permanently connected at the respective motor starters and master OFF control switches through watertight packing glands, with the other ends terminating in 3-hole plug connectors suitable for mating with and being positively locked to the respective receptacles (see 3.25.7).

3.8.2.5 Internal wiring. Internal wiring shall be installed in conduit conforming to WW-C-566 or WW-C-581, as applicable. Conduit outlets shall conform to W-C-586. A ground wire shall be carried as the fourth wire and shall be grounded to the unit in such a manner as to assure an electrical bond between all electrical equipment and the unit on which it is mounted. Branch circuit conductors, including wiring internal to starters, breakers, and other electrical wiring and devices which carry motor-running current, shall have a current-carrying capacity of not less than 125 percent of the applicable motor-running current, except that control subcircuits not carrying motor-running current shall be not smaller than No. 14 AWG wire. Such subcircuits shall include wiring internal to the control panel but not internal to starters, breakers, and other electrical devices. Plug-in type connections in branch circuits shall have a positive lock or other effective means to prevent accidental disconnection caused by vibration. Each wire in the system shall be number coded near each connector and shall be as shown on a wiring diagram which shall be furnished by the supplier.

3.9 Drives.

3.9.1 Chain drives. Unless otherwise specified herein, chains used in drive trains shall be in accordance with MIL-C-52058 and shall conform to MIL-STD-421, MIL-STD-423, and MIL-STD-424 as applicable. Chain drives shall be installed with idlers or other methods for providing adjustment.

3.9.2 V-belt drives. V-belt drives shall be in accordance with ASA B55.1, or Rubber Manufacturers Association, Inc., "Standard Specifications for Drives Using Narrow V-Belts (3V, 5V, and 8V Cross Sections)". Jack shaft drives may be used. All V-belt drives shall have provision for adjustment.

3.9.3 Gear drives. Gear drives shall be in accordance with ASA B6.6. Gear drives shall be totally enclosed and shall run in an oil bath. The enclosures shall be as specified in 3.15.4. Cast iron when used for gears shall conform to ASTM A536, analysis and heat treatment as applicable. When cast iron gears are provided, the driving pinion shall be steel.

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3.9.3.1 Torque-arm, speed-reducing gear drives. Torque-arm, speed-reducing gear drives shall conform to 3.9.2 and 3.9.3. Torque-arm, speed-reducing gear drives may be used only on those units or components of the plant that are driven by electric motors.

3.9.4 Shafting. Shafting 1-3/4 inches in diameter and larger shall be of steel in accordance with ASTM A193, type B7, hardness 25 to 32 on Rockwell "C" scale; and shall be turned, ground, and polished. The fit tolerance between all shafting and driven members shall be not more than 0.003 inch.

3.9.4.1 Keys and keyways (except for electric motors). All keys and keyways shall have the configuration shown in figure 3. Key ends shall be rounded to prevent line or point contact after installation. Keys shall be as hard as the shafting.

3.9.5 Taper-type locking devices (pulley and shaft). Pulleys shall be provided with an inner hub which clamps to the shaft by moving axially along a taper in the outer hub. The mechanical clamping shall be accomplished by the use of a screw member or members providing for mounting and removal of the pulley.

3.10 Conveyors. Belt conveyors shall have steel channel or truss frames, or equivalent construction, and may be of hinged or knockdown construction as necessary to conform to the overall dimensions for transport. The head, tail, jack, and tumbler shafts shall be mounted on antifriction bearings. Gear boxes on conveyor drives shall be equipped with bronze or antifriction bearings. Any conveyor length greater than 15 feet shall be driven at the head shaft. Each conveyor shall have a screw adjustment located at the non-powered end of the conveyor for belt tension. Adjustment shall be provided for taking up a total length of belt of at least 18 inches. Conveyor-drive pulleys shall be lagged. The head pulleys and tail pulleys shall be at least 1 inch wider than the belt and not less than 14 inches in diameter. Head pulleys shall be of solid welded construction, and tail pulleys shall be self-cleaning or of the wing type. Head and tail pulleys shall be provided with an inner hub which clamps to the shaft by moving axially along a taper in the outer hub. The mechanical clamping shall be accomplished by the use of a screw member or members. Head and tail shafts shall be not less than 2-7/16 inches in diameter. Each conveyor shall be provided with steel decking between the carrying and the return belts to prevent the spillage from falling onto the return belt and being carried into the tail pulley. Further protection shall be provided by a floating belt scraper installed to ride against the inside of the return belt near the tail pulley. Each 50-foot portable conveyor shall have a back-stop to prevent roll back of the conveyor belt.

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3.10.1 Conveyor belt. The conveyor belt shall conform to ZZ-B-206, type A, 4-ply; 30 inches in width for all conveyors, except the side delivery conveyor on jaw crusher unit No. 1 (see 3.25.3.4) which shall be 24 inches in width; and with the following exceptions. The top cover shall be 1/8 inch thick and the bottom cover shall be 1/16 inch thick with a skim coat between all plies. The tensile strength of the top and bottom covers shall be not less than 2,500 psi. The friction pull shall be not less than 16 pounds. The conveyor belt shall be provided with bolted metal plate fasteners which can be assembled and disassembled.

3.10.2 Belt carriers. All conveyor belts shall be carried on troughing rolls and return rolls.

3.10.2.1 For conveyor units No. 3, 7, 12, and 13.

3.10.2.1.1 Troughing rolls. The troughing rolls shall conform to MS16962. The troughing-roll assemblies shall be bolted to the conveyor frame and shall be spaced not more than 4 feet apart except at the foot end.

3.10.2.1.1.1 Support and troughing rolls at foot end. The first roll ahead of the tail pulley shall be a straight roll assembly conforming to MS16965, except that the support bracket may be modified to suit the application. The first roll shall be mounted under or slightly to the rear of the concentrating hopper. The next three roll assemblies shall be troughing rolls as specified in 3.10.2.1.1 except that they shall be spaced at intervals of not more than 1-1/2 feet.

3.10.2.1.2 Return rolls. The return rolls shall conform to MS16965 and shall be spaced not more than 9 feet apart.

3.10.2.2 For conveyor on jaw crusher, unit No. 1. In addition to troughing and return rolls, impact rolls shall be furnished.

3.10.2.2.1 Troughing rolls. The troughing rolls shall conform to MS16961. The assemblies shall be spaced not more than 4 feet apart and shall start from the last impact roll and continue toward the head end of the upper run.

3.10.2.2.2 Impact rolls. The area under the receiving hopper, including the area under the jaw crusher, shall be provided with impact troughing carrier rolls conforming to MS16963 mounted at intervals of not more than 1-1/2 feet.

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3.10.2.2.3 Return rolls. The return rolls and mountings shall be in accordance with MS16964 and shall be spaced not more than 9 feet apart. One return roll shall be mounted on the upper run of the conveyor ahead of the tail pulley and underneath or slightly to the rear of the back flashing on the receiving hopper. The mounting may be modified to suit the installation. If the vertical centerline of the tail pulley is within 6 inches of the back flashing on the hopper when 6 inches of belt takeup remains, the return roll shall not be required.

3.10.2.3 For conveyors on roll crusher units No. 2, 4, and 6; and feeder unit No. 9.

3.10.2.3.1 Troughing rolls. The troughing rolls shall be in accordance with MS16961. Not less than three assemblies shall be located under the receiving hopper spaced at intervals of not more than 1-1/2 feet. The balance of the assemblies shall be mounted at intervals of not more than 4 feet.

3.10.2.3.2 Return rolls. The return rolls and mountings shall be spaced not more than 9 feet apart and shall be in accordance with MS16964.

3.10.2.3.3 Return roll on upper run. One return roll shall be mounted on the upper run of the conveyor ahead of the tail pulley and underneath or slightly to the rear of the backflashing on the receiving hopper. If the vertical centerline of the tail pulley is within 6 inches of the backflashing on the hopper when 6 inches of belt takeup remains, the return roll will not be required.

3.10.3 Receiving hopper. A receiving hopper shall be located at the point on each conveyor where the materials enter upon the conveyor belt. The hopper shall be a part of the main unit structure or a separate hopper. The conveyor receiving hoppers shall be provided with sideboards wherever it is necessary to guide material onto the conveyor belts. A rubber-type skirting not less than 4 inches wide shall be provided at the lower edge of the hopper and at the sideboards to prevent spillage and to reduce wear on the belt surface.

3.10.4 Conveyor wheel oscillation. To enable the conveyor to traverse in a pivot turn centered about the foot end on the ground during stock piling of aggregates, means shall be provided to toe-in or toe-out the road wheels or to swivel the tandem assembly. Positive locks shall be provided for pivot traverse and for road travel. The locking device for road travel shall not cause weaving or side sway during towing.

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3.11 Commercial vehicular components. When specified (see 6.2), the units shall be equipped with the commercial vehicular components in accordance with 3.1.1, with table I, and as specified herein.

3.11.1 Semitrailer frame. The main frame shall extend the full length of the semitrailer. Outboard side rails shall extend rearward from the gooseneck area to the rear crossmember. Front, rear, and intermediate crossmembers shall be provided. A deck in the gooseneck area shall cover the upper fifth-wheel plate. Reinforcements for landing gear supports, jack mounts, and machinery mounts shall be provided. Frame members shall be spliced at the gooseneck area only. Where frame members are spliced, joints shall be reinforced and the spliced area shall be equal to or shall have greater strength than a continuous beam. Splices shall not cause stress concentration and shall have continuous welds to exclude moisture and rust.

3.11.1.1 Frame jacks. Both semitrailer units and cart-trailer units shall be equipped with three or four screw-type jacks, permanently attached to the chassis, of sufficient capacity to level and stabilize the unit without use of auxiliary jacks. Each jack pad shall have sufficient area to reduce ground pressure to less than 100 pound per square inch. If necessary, detachable pads may be furnished but provisions shall be made for stowing the pads on the unit in a location clearly marked as to their use.

3.11.2 Semitrailer upper fifth-wheel. The upper fifth-wheel plate shall couple to a full oscillating fifth wheel, shall be of sufficient size to cover a lower fifth wheel 36 inches in diameter, and shall conform to ICC Motor Carrier Safety Regulations 193.70(b), (e), and (h). The kingpin shall conform to the SAE J700.

3.11.3 Semitrailer landing gear. The landing gear shall be the vertical retractable two-speed type, with the legs operated and controlled by a ratchet-type handcrank. Each leg shall be capable of supporting its portion of the fully loaded semitrailer and shall be braced. Legs shall be provided with steel pads not less than 18 inches square. Supports for the crank extension shaft and clips for holding the crank, when folded, shall be provided.

3.11.4 Suspension and axles. Suspension shall be the supplier's current commercial flexible, sprung or articulated type, capable of permitting a vertical difference of not less than 8 inches between any two wheels. Axle spindles shall be heat treated and ground, and provided with tapered roller bearings. Axles and other load-carrying components shall have capacity to support the maximum rated load imposed under all operating conditions specified herein.

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3.11.5 Wheels, rims, tires, and tubes.

3.11.5.1 Wheels and rims. The semitrailer shall be equipped with dual or wide-base single wheels. The rim size shall be as recommended by the Tire and Rim Association Yearbook for the size of tire furnished.

3.11.5.2 Tires. Tires shall be tube or tubeless type with highway tread. All tires of a unit shall be of the same size and of rated capacity at least equal to the load imposed on the highest loaded tire, measured at each wheel, at the ground, with vehicle loaded to specified payload. Tires shall be of not less than 100 level quality and shall include antiozonant compounds.

3.11.5.3 Tubes. Tubes shall be heavy-duty type and of a size to suit the tires. Tire flaps shall be provided for tube-type tires in accordance with Tire and Rim Association recommendations.

3.11.6 Brakes.

3.11.6.1 Service brakes. Service brakes shall be of the full air internal expanding type and shall comply with ICC Motor Carrier Safety regulations 193.40, 193.42, 193.43, 193.45 through 193.47 and as specified herein. The brake system shall include piping, hose connections and couplings, air reservoir, slack adjusters, and all other parts required for a complete air brake system. Connections and couplings shall conform to SAE J702. All components of the braking system shall be installed to provide sufficient road clearance on rough terrain and shall be shielded from damage by striking objects. No part of the braking system shall extend below the bottom of wheel rim.

3.11.6.2 Airbrake emergency system. Unless otherwise specified (see 6.2), an approved emergency airbrake system of the spring-applied type or of the air-diaphragm mechanical-locking type, conforming to the State of California Vehicle Code, section 26310, shall be furnished. The application device shall be an integral part of, or mounted on, the service brake chamber. The brakes shall be capable of holding the fully loaded semitrailer on a 25 percent grade. In addition, at least two wood chock blocks shall be provided.

3.11.7 Lighting system. The lighting system shall be 12-volt potential and shall comply with the American Trucking Association E-3 Recommended Equipment Specification and ICC Motor Carrier Safety Regulations 193.14, 193.20, 193.22, 193.23, 193.25 through 193.29, 193.32, and 193.33. Lights and reflectors shall be recessed or guarded. The coupling receptacle shall conform to the SAE J559 and SAE J560 and shall be located in accordance with SAE J702. Connections shall be of the round type, installed at the front of the semitrailer. Directional signals shall conform to SAE J589, class A.

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3.11.8 Commercial dolly converter. When specified (see 6.2), a dolly converter shall be furnished with each semitrailer unit and shall be as specified (see 6.2).

3.11.9 Cart-trailer commercial vehicular components. Cart-trailer type units shall be equipped with a trailer lighting jumper cable and both service and emergency jumper airbrake hoses having sufficient length to connect the trailer to any towing vehicle conforming to SAE J849 and to allow correct clearance for turning. A tow bar eye in accordance with SAE J847 shall be furnished and shall be installed in accordance with SAE J849. The tow bar shall be of the A-frame type constructed and installed to provide the structural strength necessary to resist bending and twisting without permanent deformation or part failure under any operating or transport condition specified herein. The tow bar length shall meet the turning ability requirements specified in 3.7.1.3. The safety chains shall conform to ICC Safety Regulations 193.70(f). The trailer suspension, including wheels, rims, tires, and tubes, with brake systems and with a lighting system, shall conform to 3.11.4 through 3.11.7, as applicable, except that the jumpers shall be as specified herein and the articulation and sprung suspension requirements are not mandatory.

3.12 Military vehicular components. Unless otherwise specified (see 6.2), the units shall be equipped with components compatible with Military prime movers. These components shall be in accordance with 3.1.1, with table I, and as specified herein. The Military prime movers shall include the 5-ton M52 truck-tractor as shown on MS500008.

3.12.1 Semitrailer intervehicular connections. The semitrailer shall be furnished with front and rear airbrake couplings as specified in 3.12.5.1 and front and rear electrical intervehicular receptacles as specified in 3.12.6 for connecting (1) the truck-tractor airbrake and electrical systems to the semitrailer systems, and (2) the dolly airbrake and electrical systems to the semitrailer airbrake and electrical systems. The front couplings and the receptacle shall be installed on the front of the semitrailer, symmetrical about the centerline in accordance with SAE J702. Clearance about the couplings and the receptacle shall permit connection of the truck-tractor hoses and harness to the semitrailer without interference with the upper fifth-wheel plate or other components. The air brake couplings shall be protected with metal, spring, dust covers. The front electrical receptacle shall have pigtail leads terminating within the semitrailer electrical nosebox. The rear couplings and receptacle shall be installed on the rear of the semitrailer symmetrical about the centerline in accordance with SAE J849.

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3.12.2 Semitrailer frame. The semitrailer frame shall be as specified in 3.11.1, and both semitrailer and cart-trailer frame jacks shall be as specified in 3.11.1.1.

3.12.2.1 Semitrailer upper fifth-wheel plate and kingpin. The upper fifth-wheel plate shall cover a 36-inch-diameter lower fifth wheel and shall extend the entire width of the gooseneck. The plate shall be fabricated of high tensile steel with not less than 50,000 psi yield point and shall have a thickness of not less than 0.370 inch. Both longitudinal and transverse reinforcing members shall be provided. A front coupling ramp shall be furnished at the front of the plate. Ramps that are not flush with the front crossmember shall include at least four vertical gussets and shall be provided with holes for water drainage. For kingpin loads under 20,000 pounds, an SAE 2-inch-throat-diameter kingpin shall be furnished conforming to MS53036. For kingpin loads 20,000 pounds and greater, a universal kingpin having both 2- and 3-1/2-inch throats shall be furnished conforming to MS53038, MS53039, and MS53040. The kingpin shall be located as specified in table I. The center of the kingpin shall be within 1/8 inch from the longitudinal centerline of the semitrailer.

3.12.2.2 Semitrailer towing pintle hook. A 40,000-pound towing pintle assembly conforming to MS51118 shall be mounted on the rear of the semitrailer in accordance with SAE J849. Two shackles for safety chains shall be installed, one on each side of the pintle. The pintle shall provide a towing connection for the dolly converter and full trailers when the unit is being towed by a truck-tractor.

3.12.3 Semitrailer landing gear. The semitrailer shall be equipped with a retractable, folding-leg landing gear. The landing gear shall conform to MS53093 or MS53091, except that the distance between the right hand and left hand landing legs of MS53091, with corresponding changes in cross braces and shaft lengths, shall match the longitudinal side rails of the frame. The material and design of the braces shall result in rigidity at least equal to that provided by MS53091. The installation of the landing gear shall be in accordance with the landing gear manufacturer's recommendations.

3.12.4 Rear suspension. The semitrailer shall be equipped with a flexible, sprung, rear single or multiaxle suspension. The suspension system shall protect all machinery and semitrailer components from shock and vibration and shall withstand static and dynamic loads without permanent deformation or damage when the units of the plant are transported as specified herein. The sprung suspension shall be of the bar- or walking-beam-spring type having elastomer bushings and blocks to eliminate need for lubrication. The suspension

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system shall provide stops or bumpers of elastomer material to limit axle movement at full compression and at full rebound. Articulation shall provide not less than 8 inches difference in elevation between all combinations of both opposite and diagonally opposite wheels without loss of tire-to-ground contact. The suspension shall be in correct alinement with the frame and with the axles so that the diagonal measurements from the true center of the kingpin to the center of each spindle on the front axle do not differ in excess of 1/16 inch. Each subsequent axle shall be parallel to the front axle so that the axle wheelbases on each side do not differ in excess of 1/8 inch. The suspension shall be furnished with dual tire axles in accordance with the axle load limit in table I. The axles may be of the stub-spindle or through-axle type and shall conform to MS35865-C-1, except that the axles shall be equipped with straight-air-actuated foundation brakes as specified in 3.12.5 and the axle-tube lengths shall be increased approximately 20 inches while retaining the 16 inches between track width centers and the suspension mount centers on each side. Suspension mounts shall be installed on the axles in accordance with the suspension and axle manufacturer's recommendations. Tires, tubes, flaps, and wheels shall be duals as follows:

(a) Tires and flaps size 11.00 x 20 conforming to MS35388-21.
(b) Inner tubes conforming to MS35392-15.
(c) Wheels size 20 x 7.50 conforming to MS53044-6.

Each unit shall be equipped with wheel splash fenders and guards which shall restrict water and other road-surface substances thrown from the dual wheels. These components shall be in accordance with SAE J682, SAE J683, and MS51331.

3.12.5 Brake system. The semitrailer shall be equipped with a service and emergency-breakaway airbrake system complying with ICC Motor Carrier Safety Regulations and the California Vehicle Code, Section 26508, except as specified herein. The brake system shall be of the waterproof, straight-air-application type. All components of the brake system shall be located, installed, and protected to prevent damage from water, ground obstructions, and debris and to obtain maximum efficiency and ease of maintenance. The brake system shall include the following:

(a) Service and emergency line filters or air cleaners.
(b) An SAE J10, end drain reservoir with 1-1/2 times the minimum air volume recommended by SAE J813 for truck-trailers.
(c) A waterproof emergency relay valve conforming to MS53004, except that the bleed-down feature is not required.

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3.12.5.1 Brake lines. The intervehicular air lines for cart-trailers shall be furnished with couplers equipped with metal spring covers, identification tags, and coil spring hose guards on each end. The airbrake lines shall be secured to the semitrailer and cart-trailer by cushioned support clamps at intervals not exceeding 24 inches. Where the brake lines pass through metal frame members, these lines shall be either secured and protected by a grommet or shall incorporate a through-frame connector. The flexible hose shall be removable at the axle ends and the foundation brake ends. The brake lines shall be routed from first frame crossmember to the relay valve along the interior flanges of the longitudinal frame members on each side. All brake lines shall be installed to facilitate replacement from bulk supply. The brake line components shall be as specified in table II. Item quantities that are not specified shall be as required for each semitrailer and cart trailer.

Table II. Brake Line Components

Description	Requirement
Flexible air hose	MIL-H-3992, type I.
Rigid air line	SAE J844, copper tubing.
Airbrake fittings	SAE J512, airbrake type, reusable.
Intervehicular couplings	4 each, MS35746, supplied with metal spring, dust covers and color coded blue for service and red for emergency.
Identification tags	4 each, MS53007.

3.12.5.2 Foundation brake assemblies. Foundation brakes shall be installed on each wheel and shall be of the self-adjusting type. The minimum total braking swept surface area for each axle shall be not less than 560 square inches. The foundation brake friction material shall be nonfading from heat or water. The brake lining shall be sintered metallic brake lining conforming to KKK-L-370 for airbrake usage and shall be listed in the Automotive Data Book of the Friction Materials Standards Institute, Inc.

3.12.5.3 Foundation brake protection. All metal surfaces enclosed by the brakedrum shall be coated with primer conforming to TT-P-664 with the exception of (1) items constructed of materials providing inherent corrosion protection, (2) parts prime coated or painted in accordance with Government specifications, (3) the braking linings, and (4) those moving parts coated with preservative oil conforming to MIL-L-21260, grade 2.

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3.12.5.4 Chock blocks and hangers. Two wood chock blocks and two hangers shall be furnished for each cart-trailer and semitrailer. The chock blocks shall conform to figure 4. The hangers shall secure the chock blocks while not in use. Blocks and hangers shall be installed on each side of the trailer deck, forward of the wheels. The hook of each captive chain shall be snapped on a metal loop, supplied for the purpose, under the deck and the chain shall be wrapped clear of the ground and obstructions. Each captive chain shall be long enough to allow the chock and its chain to be lifted clear of the semitrailer and for the chocks to be placed between the wheels.

3.12.5.5 Emergency-parking brake system. The emergency brake system shall be of the power-applied type conforming to all legal requirements specified herein. The emergency-parking brakes shall be of the spring-applied or air-diaphragm mechanical-lock type and shall include means for manual or mechanical release when compressed air is not available. An instruction plate shall be installed on the gooseneck near the gladhands to inform personnel to release the emergency-parking brakes during prolonged production and storage periods.

3.12.6 Vehicular electrical system. The semitrailer shall be equipped with a waterproof, 24-volt, vehicular electrical system for service and blackout night travel operation. The service electrical system shall be in accordance with (1) the Electrical Equipment and Lighting section of the SAE Handbook, (2) the ATA E-3 Recommended Equipment Specification, and (3) ICC Motor Carrier Safety Regulations, except as specified herein. The blackout electrical system shall be as specified herein, using the wiring diagram shown on Drawing 8347201 for guidance. The electrical system shall be in accordance with table III. Item quantities that are not specified shall be furnished as required (AR) for semitrailer.

Table III. Vehicular Electrical System

Description	Requirement	Quantity per semitrailer
Service lights: clearance, side-marker, and identification	MS35423	13 or 15
Blackout lights; clearance side-marker	MS35424	8
Service combination lights	SAESIAT, 4-in. dia., waterproof; (SAE J759)	2
Blackout combination lights	MS51330	2
Reflectors	MS35387	8

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Table III. Vehicular Electrical System (Con't)

Description	Requirements	Quantity per semitrailer
Intervehicular harness	MIL-C-13486/12, length as specified	1
Front trailer receptacle	MS75021-1 with cover	1
Rear trailer receptacle	MS75021-2 with cover	1
Low-tension cable	MIL-C-13486	AR
Harness insulation	MIL-I-15126 and MIL-I-631	AR
Loom	SAE J562 oil proof loom	AR
Cushioned support clamps	MS21322, MS21333, and MS21334	AR
Circuit number bands	MS39020	AR
Friction connectors, waterproof type	MS27142 through 27148, and MS33800	AR
Connector lubricant	MIL-S-8660	AR
Connector holder clip	Metal, spring clip type	1 ea per connector
Threaded plugs and receptacles	Military waterproof type	AR
Terminal connectors	Spade with insulation grips (SAE J561)	AR
Circuit breakers	MS39062	3

3.12.6.1 Lighting. All lights shall be installed on the deck and shall operate as specified herein on 24 volts. Shock and vibration protection shall be provided for all lights to prolong filament life. Both clearance lights and side-marker lights shall be provided on all four corners. Identification, side-marker, and clearance lights shall be clustered and mounted on plates inserted into lamp bracket sockets conforming to SAE J725. These clusters shall be removable for storage and identified with plates (see 3.18.1) to preclude incorrect replacement for travel. The two service combination lights shall be sealed-beam units mounted in elastomer rings, or bulb, socket, and lens sealed in elastomer housings and shall be the flush or recess mounted type. The two blackout combination lights shall be installed inboard of the service combination lights. These two blackout lights shall be installed flush within 6-inch-minimum-diameter holes in the rear apron or cross member or protrude not more than 4 inches rearward and shall be protected with guards equal to 6-inch-diameter, schedule 60, steel pipe. The rear of the four

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combination lights shall be boxed and shielded to prevent damage in any environment specified hrerin. Blackout clearance and side-marker lights shall be positioned below the corresponding service clearance and side-marker lights.

3.12.6.2 Wiring. The wiring for both service and blackout lights shall be as specified in 3.12.6, except as specified herein. Grommets shall be used to prevent chafing where wire and harness pass through any member. Loom shall be furnished to protect wires and harness from road splash. Cushioned support straps and loom shall prevent whipping, chafing, or other mechanical damage to the wires and harnessed. Splices shall be insulated and waterproofed. Two or more wires following the same routing shall be formed into harnesses with either one-half overlapping turns of 0.010-inch-thick, 1-inch- wide tape or with insulation. All wires shall be number coded near each connection as shown on Drawing 8347201 and as specified herein. Friction connectors shall be lubricated with connector lubricant prior to assembly and each assembly shall be secured to the semitrailer with a connector holder clip. Waterproof electrical, semitrailer-type nosebox shall be furnished. The nosebox shall contain terminal connectors for all of the live and the two ground circuits and shall house the three circuit breakers. The circuit breakers shall be wired to protect circuits 22-460, 22-461, and 21-489. The nosebox shall be mounted on the front. From the nosebox, one main harness shall be routed rearward along the frame side member to the rear lights and incorporate branch harnesses routed to the side lights. The service combination lights shall be connected through pigtail leads to circuits 21-489 (tail) and 22-460 and 22-461 (stop and turn). The blackout combination lights shall be connected to corcuits 24 (tail) and 23 (stop). The rear receptacle shall be connected into the semitrailer system to provide lighting of other towed vehicles.

3.12.6.3 Reflectors. Reflectors shall be furnished and mounted as required by ICC regulations. In addition to reflectors required by ICC regulations, two amber reflectors shall be installed on the front corners of the trailer facing forward.

3.12.7 Front dolly converter. A dolly converter shall be furnished with each semitrailer type unit. The dolly shall permit coupling the dolly to the semitrailer by backing and shall permit cross-country travel.

3.12.7.1 Fifth wheel. Except for unit No. 8, a heavy-duty, 36-inch-plate, universal, lower fifth wheel with oscillation lockouts shall be furnished. The fifth wheel and its mechanism shall accept and couple with the 2-inch throat diameter kingpin specified in 3.12.2.1. Ramps shall be provided to guide the kingpin into the fifth wheel when backing. The height of the loaded fifth wheel shall be 56 inches plus or minus 2 inches. The mounting of the fifth wheel shall be in accordance with ICC regulations.

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3.12.7.2 Lunette and towbar frame. A lunette conforming to MS51337 shall be installed in a housing integral with the towbar frame. The lunette shall be supported the entire length of the shank. The clearance between the shank and support shall not exceed 0.035 inch. When the dolly is in towing position, the height of the lunette shall be between 34 and 36 inches from the ground line. The center of the lunette and the centerline of the towbar shall be located on the longitudinal centerline of the dolly within 1/8 inch. The dolly towbar shall be integral with the frame with single axles and hinged to the frame with tandem axles. When coupled to the semitrailer, the dolly shall turn 110 degrees both to the right and to the left of the longitudinal centerline of the semitrailer. The lunette and towbar assembly shall be compatible for towing with a Military, 5-ton, 6x6, M51 dump truck shown on MS500010 and shall permit the centerline of the truck to be at an angle of not less than 60 degrees to the right or to the left of the centerline of the dolly when in towing position. There shall be no interference between the truck and the dolly-semitrailer combination.

3.12.7.3 Landing leg. The towbar of the dolly converter shall be provided with an adjustable, folding, landing leg to permit coupling and uncoupling of the lunette and the towing pintle by one man. Ground clearance shall be not less than 20 inches when the towbar and jack are in travel position.

3.12.7.4 Dolly suspension. The dolly shall be supported on a sprung, single- or tandem-axle assembly having dual tires. The suspension components and the installation, when coupled between a truck and the unit, shall withstand towing and braking forces, cross-country articulation, and shock loads under the conditions specified herein without permanent deformation or damage. The suspension and fifth-wheel oscillation shall provide not less than 8 inches articulation between the outer duals of the dolly as measured from the level unit. The suspension and brake systems shall be as specified in 3.12.4. and 3.12.5 through 3.12.5.3. Chock blocks and the power applied emergency-parking system shall not be required.

3.12.7.5 Dolly electrical system. The electrical system of the dolly shall be as specified in 3.12.6 as applicable to dolly converters. Service and blackout rear combination lights shall be furnished and installed.

3.12.7.6 Dolly intervehicular components. The dolly shall be provided with electrical and airbrake lines to connect the truck systems to the semitrailer systems. All intervehicular components between the dolly and the truck shall extend 36 to 42 inches toward the truck from the center of the lunette eye. All intervehicular components between the dolly and the semitrailer shall be supported on a truck-tractor-type, flexible, spring pole. The pole shall be spring loaded to permit repeated bending to the horizontal

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position without damage. The spring pole shall prevent entangling or damaging the lines by either the dolly or the unit. The intervehicular component shall have a minimum length of not less than 10 feet from the dolly frame to the connections and shall allow the dolly to turn at least 110 degrees to the right and 110 degrees to the left from straight ahead. Clips, dummy couplings, and eye bolts shall be mounted on the dolly to protect and to hold the intervehicular components during storage, operation, and travel when the component is not in use. Electrical components shall be as specified in 3.12.6. Airbrake components shall conform to 3.12.5.1 and shall include hose coil spring guards on the couplings.

3.12.7.7 Safety chains. Two safety chains fitted with grab hooks shall be furnished. The chains shall extend between 36 and 42 inches beyond the center of the lunette eye. The safety chains shall be in accordance with ICC Motor Carrier Safety Regulations, except that the hooks and the safety-chain shall fit through the shackles of the towing vehicles and of the semi-trailer (see 3.12.2.2).

3.12.8 Cart-trailer Military vehicular components. Military vehicular components of the cart-trailer type units shall be in accordance with the following requirements. The length of the jumper cable, service and emergency airbrake hoses, and safety chains shall extend between 36 and 42 inches toward the towing vehicle from the center of the towbar eye. Spring clip and dummy couplings shall hold the lines when disconnected and doubled back on the towbar. These lines shall be routed to prevent damage when the tailgate of a Military truck conforming to MS500010 is dropped. The towbar eye shall conform to either MS51336 or to MS51339-1, shall be in accordance with SAE J849, and shall be installed on the towbar within 1/8 inch on either side of the longitudinal centerline of the unit. The towbar shall be of the A-frame type constructed and installed to provide the structural strength necessary to resist bending and twisting without permanent deformation or part failure under all the operating and transport conditions specified herein. The towbar length shall conform to the turning ability requirements in 3.7.1.3. The safety chains shall conform to ICC Safety Regulations and the safety chain hooks shall pass through the shackles of the Military truck-tractor (see 3.12). The cart-trailers shall be equipped with either an unsprung single 10,000-pound axle or a sprung tandem axle assembly. The axle components shall conform to MS35865-A, except that these components shall be equipped with straight-air brakes as specified in 3.12.5. Single axles shall conform to 3.12.4, except for suspension and axle. Tandem axles shall conform to 3.12.4 except for axles. Tires, tubes, flaps, and wheels shall be singles and as follows:

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(a) Tires and flaps size 11.00 x 20 conforming to MS35388-21.
(b) Tubes conforming to MS35392-15.
(c) Wheels size 20 x 7.50 conforming to MS53044-5.

The brake system and the vehicular electrical system shall be furnished in accordance with 3.12.5 through 3.12.6.3 except that the jumpers shall be as specified herein.

3.13 Fungus and moisture resistance. The electrical circuitry, including all components and connections, shall be protected from the effects of moisture and fungus growth by an overall treatment with varnish conforming to MIL-V-173 with the following exceptions:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, varnish shall be applied by spray, brush, or a combination of both to give a minimum dry film thickness of 1 mil to component and element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or functioning of the varnish.

3.13.1 Fungus-resistance treatment of conveyor belts. All conveyor belt fabric shall be treated to resist the growth of fungus.

3.14 Electromagnetic compatibility. Unless otherwise specified (see 6.2), the electromagnetic compatibility of the crushing, screening, and washing plant shall be in accordance with MIL-E-55301, tactical support class.

3.15 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.15.1 Lubricants. The plant shall operate as specified herein when lubricated with Military lubricants (see 6.3).

3.15.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II, or III. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859,

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equipped with a hydraulic coupler conforming to MS24207 and a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Lubrication fittings shall be capable of being greased by an operator while both his feet are firmly planted on a solid flat surface. Lubrication fittings shall be not more than 2 feet laterally nor more than 6 feet vertically from this solid flat surface.

3.15.3 Pressure-release device. An automatic pressure release device shall be provided where pressure lubricating equipment will damage grease seals or other parts.

3.15.4 Filling, draining, and checking provisions. Enclosures such as gear cases and transmission housings that contain a reservoir of lubricants shall be equipped with dipsticks, check plugs not less than 1/2 inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with a lubricant and for draining. The drain outlet shall be fitted with a magnetic plug, size conforming to MS35844. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the plant is in its normal operating position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to the ground. Accessibility to the drain plug, the filling means, and the lubricant-level checking device shall be obtained without removal or adjustment of accessories or parts.

3.15.5 Lubrication service.

3.15.5.1 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with grease conforming to MIL-G-10924 or MIL-G-23827 as applicable. The plant shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because Military greases are not compatible with other greases. Each component shall be tagged in a conspicuous place to indicate that Military grease has been used.

3.15.5.2 Oil lubrication. Unless otherwise preserved for packaging (see 5.2), all components shall be filled to the operating level with winter grades of Military oils designated for use in the temperature range of 0° F. to 32° F. Each lubricated component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.16 Lifting attachments. Each unit and each section (see 3.7.2.1) shall be provided with lifting attachments. The attachments shall enable the unit or section to be lifted in its normal traveling or operating position. The attachments shall be located so that not less than a 1-inch clearance is

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maintained between the slings and all exterior parts of the item being lifted. Spreader bars are permitted. Each attachment shall withstand without damage to any part of the plant not less than 2-1/2 times the normal load imposed on the attachment. Attachments for multiple slings shall be located so that the slings will converge not more than 19 feet above the lowest extremity of the load. Attachments shall be positioned so that the lifting strain will be in line with the longitudinal axis of the eye of the attachment. The inside diameter of the eye shall be not less than 3 inches. Members to which attachments are fastened and the fasteners shall withstand all stresses imposed by the attachment. Where required, structural members shall be reinforced to meet the requirements specified herein. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.17.

3.17 Tiedown attachments. Each unit and each section (see 3.7.2.1) shall be provided with integral attachments to permit tiedown to the floor or deck of the transportation medium. Each attachment shall withstand, without damage to the unit or section, its proportion of the following static loads: 8-G forward, 8-G rearward, 2-1/4-G upward, and 1-1/2-G sideward, where G equals cargo weight of the unit or section (as applicable). Attachments shall be rounded and beveled to prevent cutting of the tiedown material. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.16.

3.18 Identification marking. The plant shall be identified in accordance with MIL-STD-130. The marking shall be applied to the plant on plates conforming to MIL-P-514, type I and type V, anodized aluminum, sunken-etched or raised-relief. The location of lifting and tiedown attachments and recommended lifting and tiedown arrangements shall be included on these plates.

3.18.1 Instruction plates. Instruction plates shall be furnished and shall conform to MIL-P-514, type IX, anodized aluminum, sunken-etched or raised relief, dimensions to suit the application. These plates shall indicate sequence for start up, shut down, and normal operation; and all inherent dangers which can stem from unsafe operation or an unsafe operating position. Where applicable, the words "Warning" or "Caution" shall appear and shall be red. The shade of red shall conform to FED. STD. No. 595, No. 31136. An instruction plate with the following wording shall be placed adjacent to each high-voltage receptacle (see 3.25.7):

"WARNING
A ground rod and lead shall be installed
at power source before this plant is operated."

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3.19 Treatment and painting. The portions of the plant normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.20 Vehicle marking. The marking of each trailer-mounted unit shall be in accordance with MIL-STD-642. The gross vehicle weight of each trailer-mounted unit and the weight of each sectionalized component shall be marked on the sides in such a manner at to be discernible. The prescribed tire pressure shall be marked near the tires. Lifting and tiedown attachments shall be marked "LIFT HERE" or "TIEDOWN HERE" as applicable. Marking of lifting and tiedown attachments for sections shall include the word "SECTIONALIZATION."

3.21 Technical publications to accompany crusher. Such technical publications as are specified shall be furnished (see 6.2).

3.22 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished (see 6.2).

3.23 Toolbox. All units, except units No. 5, No. 11, No. 14, and No. 15 shall be provided with a toolbox made of sheet steel not less than 0.070 inch thick. The toolbox shall have a hinged lid and two hook-type tension fasteners to secure the lid. The lid shall open at least 90 degrees and shall be held open by gravity or detent braces. The toolbox shall be securely fastened to the unit in a protected position and shall be of sufficient size to hold the tools specified in 3.22 and the documents specified in 3.21 applicable to the unit. The toolboxes shall be accessible from the ground level.

3.24 Items to be furnished with plants. No shipment of plants shall be made unless repair parts, maintenance tools, and accessories are included with the shipment or unless approval for shipment of the plants without such items has been recieved from the contracting officer.

3.25 Capacity and unit requirements.

3.25.1 Capacity. When processing rock as specified in 3.25.1.1 which requires not less than 80 percent crushing by jaw crusher unit No. 1 and containing 28-inch by 38-inch maximum size rock, the plant shall have the capacities specified in 3.25.1.2, 3.25.1.3, and 3.25.1.4.

3.25.1.1 Quarry rock. The quarry rock shall be diabase or traprock having an average compressive strength of not less than 32,000 pounds per square inch (psi) when tested in accordance with ASTM C170 and shall have a percentage of

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wear not exceeding 20 when tested in accordance with SS-R-406, method 208.11.

3.25.1.2 Washed material capacity. When the material being fed contains not less than 8 percent CL soil as defined in MIL-STD-619 with a moisture content above the plastic limit, the plant shall produce an average of not less than 225 tph. Not less than 40 percent of the end product shall be fine aggregate (see 3.25.1.2.1) and the remainder shall be coarse aggregate (see 3.25.1.2.2).

3.25.1.2.1 Fine aggregate. The fine aggregate produced shall be 1/4-inch minus within the following tolerances:

(a) Not more than 10 percent shall be retained on the 1/4-inch sieve, nor more than 10 percent shall pass the No. 100 mesh sieve.
(b) Not more than 1 percent shall be minus No. 200 mesh when tested in accordance with SS-R-406, method 202-11.

3.25.1.2.2 Coarse aggregate. The coarse aggregate produced shall be 1-1/2-inch to 1/4-inch material within the following tolerances:

(a) No more than 5 percent shall be retained on the 1-1/2-inch square screen openings nor more than 10 percent shall pass the 1/4-inch sieve.
(b) Not more than 0.5 percent clay lumps when tested in accordance with SS-R-406, method 205.0.

3.25.1.3 Dry material capacity. The plant shall process an average of not less than 125 tph with not less than 65 percent of the end product being fine aggregate (see 3.25.1.3.1) and the remainder coarse aggregate (see 25.1.3.2).

3.25.1.3.1 Fine aggregate. The fine aggregate produced shall be 1/4-inch minus with not more than 5 percent retained on the 1/4-inch mesh sieve.

3.25.1.3.2 Coarse aggregate. The coarse aggregate produced shall be 3/4-inch to 1/4-inch material within the following tolerance: Not more than 5 percent shall be retained on the 3/4-inch square screen openings nor more than 10 percent shall pass the 1/4-inch mesh sieve.

3.25.1.4 Dry material gross capacity. The plant shall produce an average of not less than 300 tph of 1-1/2-inch minus material with not more than 5 percent retained on the 1-1/2-inch square screen openings.

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3.25.2 Flow of material. Material delivered from a pit or quarry to the jaw-type crusher (unit No. 1) shall be crushed for maximum size and then delivered to the roll-type crusher (unit No. 2). Material shall be further reduced in unit No. 2, fines rejected if necessary, and conveyed by the transfer belt conveyor (unit No. 3) to the roll-type screen and washer crusher (unit No. 4) for washing, screening to size, and crushing of oversize material. The sized and washed coarse material shall then be conveyed by the transfer belt conveyor (unit No. 12) to a stock pile, or by the conveyor of the riciprocating-plate feeder (unit No. 9) to the twin-paddle-type scrubber and washer (unit No. 10) for further scrubbing and cleaning and thence by conveyor unit No. 12 to a stock pile. The fine materials that pass through the sizing and sand screen on unit No. 4 shall pass into the settling chamber of the twin-screw dehydrator (unit No. 8) for delivery by means of the twin-inclined screws to the transfer belt conveyor (unit No. 13) for stock piling. The oversize material shall be crushed by the rolls on unit No. 4 and shall be returned by conveyor system to unit No. 4 for washing, sizing, and screening. If additional fines are needed the oversize crushed material from unit No. 4 shall be conveyed to the roll-type crusher (unit No. 6) for crushing and screening; and the materials from unit No. 4 which pass the first deck of the shaker screen but do not pass the sand screen may have a controlled amount conveyed to the roll crusher on unit No. 6 for making additional fines. The material which passes the sand screen on unit No. 6 shall be conveyed to the settling chamber on unit No. 4 and the balance of the material shall be crushed and delivered by the belt conveyor (unit No. 7) to conveyor unit No. 3 which shall deliver the material to unit No. 4 for washing, sizing, and screening to complete the circuit.

3.25.3 Unit No. 1, crusher, jaw type. The jaw crusher unit shall consist of a chassis upon which shall be mounted an apron feeder with material-receiving hopper; vibrating grizzly with bypass chute, bypass hopper, and side-delivery conveyor; a jaw crusher; an end delivery conveyor; a diesel power unit; and all auxiliary fastenings, mountings, drives, guards, walkway, and clutch. The jaw crusher shall crush the material specified in 3.25.1 from a size of 28 by 38 inches down to a size that will pass a 4-inch opening and shall deliver the finished product to the feeder conveyor of the secondary crusher (unit No. 2).

3.25.3.1 Chassis. The chassis shall be of the semitrailer type (see 3.11 or 3.12).

3.25.3.2 Apron feeder. The crusher shall be fed by an apron feeder. The feeder shall be constructed with slats made of forged steel plates not less than 42 inches long, 9 inches wide, and 1/2 inch thick. The slats shall be mounted on three steel roller chains of not less than 6 inches pitch.

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Each chain link shall have attachments for not less than two bolts not less than 5/8 inch in diameter for attaching slats to the chain. The end of the slats shall have flanges not less than 5/16 inch wide which shall overlap from one slat to the other and prevent spillage on the side of the apron-feeder. The overall length of the apron-feeder carrying slats, when mounted on the chain, shall be not less than 14 feet, measured from the extreme receiving end to the extreme delivery end. The head shaft of the feeder shall be not less than 3-7/16 inches in diameter and shall mount two steel sprockets of not less than seven teeth with appropriate pitch to drive the two outer chains on which the apron feeder slats are bolted. The tail shaft shall be not less than 3-7/16 inches in diameter. There shall be two sprockets on the tail shaft, matching those on the head shaft, on which the two outer chains of the slats shall ride. Supporting the slats on the top run of the feeder, there shall be not less than seven idler shafts of not less than 2-7/16-inch diameter. On each shaft there shall be three idler rollers not less than 9 inches in diameter on which the slat chain runs. The bearings for these idler shafts shall be the self-alining ball-and-socket type. The feeder shall be equipped with a steel hopper. The sides of the hopper shall extend vertically from the slats to a height of 6 inches and then slope outward at an angle of 45 degrees. The hopper sheets shall be not less than 3/8 inch in nominal thickness and shall have additional wear plates. The receiving lip of the hopper shall slope back at a 45-degree angle from the feeder. The hopper shall be not less than 8 feet wide and shall be equipped with structural steel framing for extending the width to not less than 12 feet, using planking not less than 3 inches wide. The sides of the hopper shall be extended into the jaw crusher hopper. The main frame of the apron-feeder shall be not less than 10-inch by 25-pound-per-foot-channel steel and shall be rigidly X-braced and cross braced. The apron-feeder frame shall be pivotly mounted with two permanently installed hydraulic rams on the delivery end for the purpose of lowering the traveling height of the feeder during transport. The drive for the apron-feeder shall be from the crusher by a drive as specified in 3.9. A reduction unit shall be built into the feeder which will give an apron-slat speed of not less than 12 fpm. A friction-type band clutch shall be provided in the drive that will stop and start the feeder. The clutch shall have a control lever mounted on the operator's platform. All bearings shall be fitted with grease fittings.

3.25.3.3 Vibrating grizzly. The unit shall be equipped with a vibrating grizzly not less than 42 inches wide and not less than 5 feet long having two decks and equipped with bypass chutes. The grizzly shall be driven from the crusher by a drive as specified in 3.9. The top deck shall consist of heavy-duty, high-carbon-steel bars or rails (extending the full length of the grizzly)

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which shall withstand the discharge of heavy abrasive rock from the apron feeder without damage or permanent deformation. The bars or rails shall be spaced to permit passing 4-inch minus rock onto the lower deck. The second deck shall be equipped with a 3/4-inch slotted screen cloth and shall screen off one size and shall discharge the oversize onto the end delivery conveyor belt. A hopper shall be provided below the second deck and shall contain a flop gate or means of bypassing material either from the hopper to the end delivery conveyor or into the side-delivery conveyor.

3.25.3.4 Side-delivery conveyor. The side-delivery conveyor shall be driven by a drive as specified in 3.9. The conveyor belt shall be not less than 24 inches wide. The discharge ground clearance shall be not less than 12 feet with the conveyor elevated to an angle of not more than 18 degrees. Belt carriers shall be as specified in 3.10.2.2, except that impact rolls shall not be furnished.

3.25.3.5 Jaw crusher. The crusher shall be of the jaw type with overhead eccentric action. The opening of the jaws measured at the top shall be not less than 30 by 40 inches at nominal 4-inch closed setting. The jaws shall be adjustable for a closed setting from a minimum of not more than 4 inches to not less than 8 inches. The jaw crusher unit shall be powered by an integrally mounted diesel engine which shall conform to 3.8.1. The jaw crusher shall crush materials ranging in size from a cross section having length and width dimensions 2 inches less than the top opening dimensions of the crusher jaws down to the size of the minimum jaw setting. The maximum size rock for a 30- by 40-inch crusher shall be 28 by 38 inches.

3.25.3.5.1 Frame. The jaw crusher frame shall be constructed of cast steel or welded steel plate. Welded steel plate shall be stress relieved after welding in accordance with the Standard Qualification Procedure of the American Welding Society for weldments of this size. Cast steel, when used, shall be in accordance with ASTM A148, with analysis and heat treatment as recommended by the supplier of the jaw crusher. The contact points on the frame which support the parts that transmit the crushing stresses to the frame shall be machined, or provisions shall be made to distribute the crushing load on the frame so that the frame will withstand the stresses imposed when crushing maximum size material without malfunction.

3.25.3.5.2 Jaw and cheek plates. Jaw and cheek plates shall be made of manganese steel in accordance with ASTM A128, with analysis and heat treatment as recommended by the supplier of the jaw crusher. The movable and stationary jaw plates shall be reversible and shall be secured so that they will not loosen and shift during the crushing operation. The back surfaces

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of the jaw plates shall be in a plane so that they will fit against their respective supporting member and provide equal load pressures in operation. The jaw plates shall be of the heavy-duty type and shall be not less than 4 inches thick at the main crushing area measured from the back bearing surface of the plate to the valley of the corrugations and shall be not less than 2-1/2 inches thick at the top and bottom of the plate measured in the same manner. Corrugations shall be essentially a sinusoidal configuration and shall be not less than 3/4 inch deep measured from crest to valley. The jaw plates shall be removable for replacement without removal of other accessories or components except cheek plates and wedges. The jaw plates shall have inserts for attaching lifting eyes or similar means for removing or inserting new jaw plates. Cheek plates shall be mounted on the frame between both sides of the jaw plates to protect the frame during the crushing operation. The cheek plates shall be constructed of two or more replaceable pieces and shall not loosen and shift during the crushing operation.

3.25.3.5.3 Pitman. The pitman shall be made of cast or fabricated steel reinforced with ribs and shall be machined to receive the movable jaw plate. Cast steel when used shall be in accordance with ASTM A148 with analysis and heat treatment as recommended by the supplier of the jaw crusher. Fabricated steel shall be stress relieved in accordance with the Standard Qualification Procedure of the American Welding Society for weldments of this size. At the lower, or toggle, end of the pitman, provision shall be made to receive a renewable toggle seat.

3.25.3.5.4 Pitman shaft bearings. The pitman shaft shall be mounted on antifriction bearings. The bearings shall be protected by labyrinth-type grease and dust seals which shall prevent dust and other foreign matter from entering the bearings. The bearings shall conform to FF-B-185 or FF-B-187, except that taper bore bearings and taper sleeve dapters will be permitted.

3.25.3.5.4 Pitman shaft bearings. The pitman and pitman shaft shall be mounted on antifriction bearings. The bearings shall be protected by labyrinth-type grease and dust seals which shall prevent dust and other foreign matter from entering the bearings. The bearings shall conform to FF-B-185 or FF-B-187, except that taper bore bearings and taper sleeve adapters will be permitted.

3.25.3.5.6 Flywheels. Two malleable iron flywheels conforming to ASTM A47 shall be mounted on the pitman shaft, shall be counter-balanced to offset the mass of the eccentric revolving parts, and shall be of sufficient size to provide inertia to transmit and equalize the energy necessary for crushing maximum size rocks without varying the speed more than 10 percent. One flywheel shall be machined for V-belts, and used as a driving pulley. The drive shall be as specified in 3.9.2.

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3.25.3.5.7 Toggle plate. A cast iron or steel toggle plate shall be provided to rest between the toggle seat on the pitman and the seat in the toggle wedge of the adjusting mechanism. The toggle plate shall be held in place by means of a tension bolt and shall be spring arranged so that the bolt will not bind or bend during operation and adjustment. The toggle plate shall be constructed so that it will break or deform when uncrushable material such as tramp iron enters the jaw crusher and shall fail before any other part of the crushing unit is stressed beyond the safe working stress of the steel from which it is made, or an equivalent safety arrangement shall be provided to protect the jaw crusher from excessive stresses. The toggle plate shall not break or permanently deform when crushing rock as specified in 3.25.1 and 3.25.3. The toggle plate shall be accessible for replacements without the removal of other components of the jaw crusher.

3.25.3.5.8 Adjusting mechanism. An adjusting mechanism of the wedge-and-screw or shim type shall be provided which will permit adjustment of the opening between the jaws. The adjusting mechanism shall be cast steel or fabricated steel plate and shall be reinforced to withstand crushing of maximum size material. All contact surfaces of the adjusting mechanism shall be machined to insure uniform distribution of the load on the respective parts of the jaw crusher frame. A stop shall be provided to prevent adjustment to actual contact of the jaws during any part of the crushing cycle.

3.25.3.5.9 Jaw setting. The jaw setting shall be measured on the closed side and shall be the greatest opening in the maximum discharge position of the jaws in the crushing cycle. Measurement for the jaw opening shall be along or parallel to the longitudinal axis of the unit on a horizontal plane. Variation between any two measurements for any jaw setting shall be not greater than 1/4 inch.

3.25.3.6 End delivery conveyor. The end delivery conveyor shall be of channel-type construction. The conveyor shall receive material from the grizzly chutes and jaw crusher and deliver the crushed and bypass material beyond the front of the trailer chassis to the hopper of the feeder conveyor on the secondary crusher (unit No. 2). The conveyor shall be driven by a drive as specified in 3.9 to the head shaft at the delivery end. Belt carriers shall be as specified in 3.10.2.2.

3.25.3.7 Power unit for unit No. 1. The power unit shall drive all the units on the jaw-type crusher and shall be mounted above the delivery conveyor and supported by means of a heavy frame attached to the main chassis of the unit.

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3.25.4 Unit No. 2, crusher, roll-type, secondary. The roll-type crusher unit for secondary crushing shall consist of a chassis upon which are mounted a feeder conveyor, a vibrator screen of the inclined or horizontal type, a roll-crusher, a fines-rejection conveyor, a main-delivery conveyor, a power unit, and the necessary hoppers, spouts, drives, guards, catwalks, ladders, mountings, and connections to make a complete unit. The unit shall be capable of crushing material from a size passing a grating with openings not greater than 6 by 8 inches square to a size 100 percent of which will pass 2-inch square openings and shall be capable of crushing material of 4- by 6-inch opening size to pass 1-1/2-inch openings. The materials received by the feeder conveyor shall be delivered to the top deck of the vibrator screen. The materials passing over the top deck shall be directed into the roll crusher. The materials passing through the top deck and, when desired, materials passing through the bottom deck shall be spouted to the delivery conveyor located longitudinally underneath the screen and roll crusher. The materials passing through the roll crusher shall be delivered to the delivery conveyor underneath, thence to the crusher (unit No. 4) by the portable belt conveyor (unit No. 3). When fines from the lower deck screen are to be rejected, the fines-rejection conveyor shall be set up in place and driven by an electric motor. The fines-rejection conveyor shall discharge the fines to the ground or to a truck.

3.25.4.1 Chassis. The chassis shall be of the semitrailer type (see 3.11 or 3.12).

3.25.4.2 Feeder conveyor. The feeder conveyor shall be mounted over the vibrator screen and roll crusher and shall be supported from the main frame of the chassis. The feeder-conveyor frame shall be of steel channel construction and the lower end of the feeder conveyor shall be hinged and provided with a worm-and-gear-type winch, or equivalent mechanism, for folding the conveyor while the unit is in transit. The conveyor shall be driven by a drive as specified in 3.9, shall be powered by an electric motor having a rating of not less than 10 horsepower, and shall be provided with a reduction drive and back-stop to prevent roll-back of the conveyor belt. The motor control switch shall be accessible for operation from ground level. Belt carriers shall be as specified in 3.10.2.3.

3.25.4.3 Vibrator screen. The vibrator screen shall be of the inclined, positive-throw or horizontal vibrator type, with four decks. The top deck shall be used for oversize material, the second deck for sized material, and the third deck for rejecting fines. A fourth deck may be used for spouting rejected fines to the fines-rejection conveyor. Wire screen shall be as specified in 3.4.3 and deck lengths shall be as follows: top - 12 feet;

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second - 12 feet; third - 10 feet; fourth - 6 feet. The live frame of the screen with attached screen cloths shall be balanced so that material will flow evenly across the full width of the screen and shall be capable of operating at a maximum speed of 1,200 rpm without causing the supporting framework to vibrate. The live frame of the screen shall be provided with supporting rubbers of springs attached to the live frame or supporting frame and shall receive its motion from a balanced eccentric shaft. The eccentric drive shaft shall be supported by antifriction bearings mounted in housings with grease-sealed closures. The live frame of the vibrator, to which the screen cloths shall be fastened, shall be constructed of steel plates or structural-member sides securely held to original width by steel bars, structural members, or pipes bolted or welded to the live-frame sides. Provision shall be made for supporting screen cloths longitudinally and providing a rubber cushion between screen cloths and longitudinal supports. The longitudinal supports shall be located so as to create a crown in the screen cloth of not less than 1/2 inch. Provision shall also be made for changing screen cloths using common tools. The three decks of the vibrator screen shall be provided with screen cloths having 2-inch-square openings, 5/8-inch-square openings, and 1/4- by 1-inch slotted openings, respectively. The drive shaft of the screen shall be driven by a drive as specified in 3.9, shall not break or permanently deform with a 50-percent overload on the screen, and shall be powered by an electric motor rated at not less than 15 horsepower. The motor shall be mounted on the frame of the unit with motor controls accessible for operation from the ground level. The stationary frame of the vibrator shall be constructed of structural members or castings and shall be securely mounted. If inclined screens are used, provision shall be made to raise or lower the stationary frame of the vibrator screen by means of hydraulic jacks, winch, or equivalent means so that the screen can be lowered for transit, if necessary, and raised into operating position.

3.25.4.4 Roll crusher. The roll crusher shall consist of two detachable roll shells mounted on cast-steel spiders supported on shafts of molybdenum or nickel-chromium steel or equal. The shafts shall be not less than 9 inches in diameter at the bearings, mounted in antifriction bearings at each end, and supported in cast-iron or steel housings mounted between heavy guide bars on the main chassis frame. The roll shells shall consist of crushing-surface steel castings not less than 54 inches in outside diameter with not less than 24-inch coarse corrugated face and shall be not less than 2 inches thick. The inside and outside diameter of the shells shall be concentric within 3/32 inch. The roll shells shall be mounted so that the maximum opening between shells will be not over 3/16 inch when both shells touch at one point during one revolution and shall rotate at a peripheral speed of 400 to 575 feet per minute. The shells shall be secured to the spiders by wedge-type circular segments, or similar means, so that the shells will not slip or shift and change the initial setting while in operation. When wedge-type

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circular segments are used, the spider, the wedge-type circular segments, and the shells shall be machined to provide a tapered seat for mounting the shell, and the wedge-type circular segments shall be held in place by through bolts. The spiders shall be keyed and securely held on the shaft so that the spiders will not shift while in operation. The spiders may be of the web or spoke type of one-piece or multiple-piece construction. The rolls shall be driven and timed by star gears. The teeth on the star gears shall be of sufficient length to permit adjustment of the sliding roll shaft so that the space between the two rolls may be varied over a total range of not less than 5 inches including a negative adjustment of 1 inch. The star gears shall be lubricated and shall be enclosed in a dustproof housing. The stationary roll shaft shall be driven from the main drive shaft by means of a spur gear and pinion. The spur gear and pinion shall be cast or forged steel, with cut teeth, and shall be lubricated and enclosed in a dustproof housing. The face and pitch of teeth of the spur gear shall be of sufficient size to transmit the full horsepower of the engine without exceeding the stress limitations specified in 3.4.1 on the gear teeth when rock of maximum size is being crushed. The main drive shaft shall be mounted in antifriction bearing pillow blocks bolted to the frame of the crusher rolls or supporting frame for rolls. Provision shall be made for longitudinal expansion of the drive and roll shafts. The main-drive shaft shall be driven from a power unit mounted on the same chassis by a V-belt drive as specified in 3.9.2. The stationary roll-shaft bearing housings shall be bolted in place on the roll frame or mounted between stationary stop blocks and shall be alined square with the roll frame. The sliding movable roll-shaft bearing housing shall be mounted on the roll frame between an adjusting block or take-up bolt and a spring-bolt equalizer block for adjusting the opening between rolls. The sliding roll-shaft housing shall be kept in position against the adjusting block or takeup bolt either (1) by means of cages containing springs of sufficient strength to provide the necessary crushing force for crushing rock and have sufficient flexibility to permit foreign metallic bodies to pass the rolls without injury to the driving mechanism; or (2) provisions shall be made to protect the driving mechanism by other safety devices. A hopper shall be provided over the rolls for receiving the material fed to the rolls. The hopper shall be constructed so that the material will be delivered onto the stationary roll at a point between 15 and 20 degrees on the far side of a vertical line drawn through the center of the stationary roll shell to provide the maximum time for spreading the material over the roll surface before it comes in contact with the sliding roll. Cheek plates of manganese or abrasion-resistant steel shall be provided to prevent spilling of material and to minimize built-up edges on shells by allowing crushing to the extreme edges of the shells. Cheek plates shall be renewable and arranged to vary the clearance between cheek plates and edge of shell. The spring and roll adjustment shall be accessible for adjustment without dismantling the crusher.

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3.25.4.5 Fines-rejection conveyor. A fines-rejection conveyor shall be provided with a one-piece or sectional channel-steel frame 15 feet long plus or minus 1 foot. The conveyor shall be driven by a drive as specified in 3.9, powered by means of an electric motor rated at not less than 5 horsepower, and shall be provided with a reduction drive. The motor control switch shall be accessible from the ground level. The conveyor shall be not less than 13 feet long, center to center of the terminal pulleys, and shall be located at the side of the plant. The conveyor shall be supported at the lower end by the frame of the crusher and suspended at the upper end from braces attached to the superstructure of the crusher. The conveyor shall be capable of being removed or folded when the unit is in transit. The fines-rejection conveyor shall be provided with a hopper built into the main structure, with a damper so arranged that fines may be rejected over the fines-rejection conveyor to the side of the plant or combined with the finished product on the main-delivery conveyor. Belt carriers shall be as specified in 3.10.2.3.

3.25.4.6 Main-delivery conveyor. The main-delivery conveyor shall be mounted underneath the roll crusher so as to receive all materials crushed and spilled by the roll shells, together with materials passing the top deck, second deck, and third deck, and to deliver them beyond the front of the main chassis to the hopper on the portable belt conveyor (unit No. 3) for delivery to the secondary crusher (unit No. 4). The conveyor shall be provided with a channel-type steel frame. The conveyor shall be driven by a drive as specified in 3.9, powered by means of an electric motor rated at not less than 5 horsepower, and shall be provided with a reduction drive. The motor control switch shall be accessible for operation from the ground. Belt carriers shall be as specified in 3.10.2.3.

3.25.4.7 Power unit for unit No. 2. The power unit for driving the roll-type secondary crusher unit shall be mounted above the main-delivery conveyor and supported by means of a heavy frame attached to the main chassis of the unit. The power unit shall be mounted on sliding plates and shall have longitudinal screws for adjusting the belt tension. The fuel tank shall be mounted on the main chassis and shall be connected to the engine fuel pump by means of copper tubing. The power shall be transmitted by a V-belt drive conforming to 3.9.2 to the main drive shaft of the crusher.

3.25.5 Unit No. 3, conveyor, belt. The belt conveyor shall receive the crushed rock and fines from the longitudinal main-delivery conveyor on the roll crusher (unit No. 2) and deliver them into the receiving chamber of the combination scrubber and scalper screen on the roll-type screen and washer crusher (unit No. 4), when the plant is operated as a washing plant, or deliver the same materials into the receiving end of the horizontal gradation screen on unit No. 4 when the gradation screen is operated in reversed direction

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under dry conditions. The belt conveyor shall consist of a hydraulic cradle chassis equipped with supporting and hydraulic raising mechanism, upon which shall be mounted a 30-inch-wide by 50-foot-long conveyor, an electric motor rated at not less than 15 horsepower and a receiving hopper. The motor-control switch shall be mounted on the conveyor frame and shall be accessible for operation from the ground level. The unit shall be driven by a drive as specified in 3.9 and provided with an arrangement to permit change of conveyor speeds between 300 fpm and 400 fpm.

3.25.5.1 Chassis. The hydraulic cradle chassis shall be capable of raising the 50-foot conveyor to a discharge height of at least 16 feet and shall lower to a height not to exceed 8 feet 6 inches for transit. Vehicular components shall be in accordance with the cart-trailer requirements (see 3.11.9 or 3.12.8). The chassis shall be equipped with a raising device consisting of two hydraulic pressure rams mounted in supporting pipes of the chassis. The oil, under pressure, shall be delivered by a hand pump to the rams. The chassis shall be so constructed that the raising mechanism and forces due to raising are confined to the chassis, and not transmitted to the conveyor or through the conveyor truss frame. The chassis shall be constructed so that one man can raise or lower the conveyor to an operating or transit position. Safety locks shall be installed on both rams as well as additional hand-operated valves in each hydraulic line to the rams.

3.25.5.2 Conveyor. The conveyor shall be not less than 50 feet long overall, frame end to frame end, of knock-down truss frame or equivalent construction, and shall be divided into three or more sections to facilitate shipping and to decrease the specified length of the conveyor. Belt carriers shall be as specified in 3.10.2.1.

3.25.6 Unit No. 4, crusher, screen and washer, roll-type. The roll-type, screen and washer crusher shall consist of a chassis upon which are mounted a combination revolving scrubber and scalping screen, a horizontal reversible double-deck gradation screen, a roll-type reduction crusher, a system of water piping, a crushed-rock delivery conveyor, a detachable return conveyor, a power unit and all the necessary combinations of drives, guards, ladders, catwalks, spouts, hoppers, fines-settling chamber, flumes, pipes, pressure hose, valves, and all attachments necessary to make a complete roll-type screen and washer crusher. The crusher unit shall be constructed so that the materials delivered by the belt conveyor (unit No. 3) to the receiving end of a combination revolving scrubber and scalping screen shall be subjected to a preliminary spray of water and thorough saturation, after which the material shall be delivered by means of the spiral vanes of the scrubber to the revolving scalping screen for preliminary mixing or scalping while being subjected to water spray throughout the length of the screen. The materials, together with the water passing through the perforations of the revolving scalping screen, shall be delivered by a revolving spout or drum to the

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receiving end of the top deck of a horizontal sizing or gradation screen when it is set to direct the flow of material opposite to the flow traveling through the scalping screen. The materials that do not pass through the perforations of the scalping screen shall be delivered to the hopper of the roll-type reduction crusher and then to a detachable return conveyor by means of a crushed-rock delivery conveyor below the rolls for recirculation through the revolving scrubber drum and screen. The materials that pass over the top deck of the horizontal gradation screen shall be delivered to the hopper of the belt conveyor (unit No. 12), and then directly to a coarse-rock stock pile. The materials passing through the perforations in the top deck on the horizontal screen but retained on the bottom deck shall be delivered to a pea-rock spout and thence to the ground as waste, or spouted to combine with the coarse material on the portable rock conveyor (unit No. 12), all or in part as determined by adjustment of a pea-rock slide valve to suit requirements for fine or coarse material. Pea-rock may also be delivered to unit No. 6 for further reduction by arranging the pea-rock spout to empty onto the side-feeder conveyor of unit No. 6 as specified in 3.25.8.6. The materials, together with water, which pass through the bottom deck of the horizontal gradation screen shall be delivered into a settling chamber located underneath and lengthwise of the horizontal gradation screen. The materials that settle to the bottom of the settling chamber shall be conveyed upward and to the side of the unit by means of the twin screws of the dehydrator (unit No. 8), for dewatering and delivering to the belt conveyor (unit No. 7), and then to a fines stock pile. The water from the settling chamber shall be delivered over weirs to a central flume located at the side of the unit and flow through portable flumes to a waste ditch or sump.

3.25.6.1 Chassis. The chassis of the unit shall be of the semitrailer type (see 3.11 or 3.12).

3.25.6.2 Revolving scrubber and scalping screen. The combination revolving scrubber and scalping screen shall consist of a perforated plate screen 60 inches plus or minus 1 inch in diameter by 16 feet plus or minus 6 inches long with 1-1/2-inch plus or minus 1/16 inch square perforations, and with a scrubber drum not less than 72 inches in diameter by 4 feet long attached to one end of the screen. The scrubber drum shall be constructed of steel plates not less than 1/4 inch in thickness, lined with removable steel liner plates not less than 3/8 inch in thickness, and shall be equipped with not less than six diagonal baffles or vanes not less than 3/8 inch in thickness arranged to propel the materials away from the receiving end and deliver them inside the revolving scalper screen. The circumferential space between the screen and drum shall be closed by a plate with liners so that the scrubber drum forms a cylindrical reservoir 6 inches plus or minus 2 inches deep where

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it is attached to the screen, to receive and agitate materials before delivering them to the screen. Outside the revolving screen there shall be a revolving plate drum 72 inches plus or minus 2 inches in diameter by 12 feet plus or minus 6 inches long for delivering the material and water passing through the revolving screen to a gradation or sizing screen. The outside plate drum may be a continuation of the scrubber drum but without vanes. Spacers shall be provided between the outside drum and screen to keep the screen and drum concentric about the same longitudinal axis. A spray pipe not less than 5 inches in diameter, fastened with brackets at each end, shall pass through the entire length of the revolving screen. The spray pipe shall have openings for spray nozzles located 4 inches plus or minus 1/2 inch apart for producing a spray of water throughout the length of the revolving screen and scrubber drum. The spray pipe shall receive water from a 6-inch header water inlet pipe, or by other means, mounted on the plant, and shall have a valve for regulating the flow of water. The connection between the header pipe on the unit and the pump line shall be made by a 6-inch rubber pressure hose conforming to ZZ-H-561, grade B, type I, 16 feet long, with flanges at each end. Machined steel tires, which shall travel on supporting trunnion rollers made of abrasion-resistant steel, shall be fastened to the outside of the drum. The combination scrubber and screen shall be driven from the power unit of unit No. 4 through transmission and clutch control by friction-trunnion-roller drive or a drive as specified in 3.9. If friction trunnion drive is used, the trunnion rollers and tires around the drum shall be accurately machined so that there will be a tolerance not over 1/32 inch in circumference on the running surface of the pair of tires. The friction trunnion rollers shall be shrunk or press-fitted and keyed to shafts which shall revolve in antifriction pillow blocks at each end. If spur or bevel gear, or chain drive is used at the ends, or around the drum for drive, and tires are rough-machined, the trunnion rollers shall be mounted on separate shafts, revolving in antifriction or bronze-bushed pillow blocks, or the trunnion rollers may be mounted on two shafts, with provision made to idle one trunnion roller on each shaft. The revolving scrubber drum and screen shall be supported by a separate structural-steel subframe or by the steel frame of the plant, shall be set at a slope of 1-1/4 inches to the foot with the horizontal, and shall be kept in position by means of a thrust roller not less than 8 inches in diameter mounted on a vertical shaft running on antifriction bearings with dustproof and waterproof housings. A separate hopper shall be provided at the end of the scrubber drum and screen to gather the waste materials and water from the drum, deliver them to the gradation screen, receive the oversize materials from the revolving screen, and deliver them to the crushing rolls on the unit.

3.25.6.3 Gradation or sizing screen. The gradation or sizing screen shall be of the horizontal, reversible, double-deck-shaker, or swinging type so that the material received from the revolving scrubber and screen can flow

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over the gradation screen in one direction when washed material is being screened; or in the opposite direction when dry material received directly from conveyors, bypassing the revolving scrubber and screen, is being screened. The horizontal gradation screen shall consist of a pan 48 inches, plus or minus 6 inches wide, 16 to 17 feet long, and not less than 18 inches deep, with removable ends for adapting the screen pan for travel of material in either direction as desired, and shall support two decks of screens. The screens may be made of perforated plate or steel wire. The wire or perforated-plate screens shall be made up in panels 48 inches wide by 60 inches long. Wire screen shall be as specified in 3.4.3. The top-deck screen shall have 1/2-inch-square openings and the bottom-deck screen shall have 1/4-inch-square openings. A slide gate for pea rock shall be provided at the delivery end of the bottom deck to vary the amount of pea rock, as desired, for delivery through a pea-rock spout. Not less than three transverse spray pipes above the top deck of the gradation screen shall be provided for receiving water under pressure from the main water header on the unit. The spray pipes shall be provided with valves for regulating the flow of the water. The gradation screen shall be supported by not less than three cross shafts. Each cross shaft shall be not less than 2 inches in diameter and shall swing in antifriction bearings from cast-iron or semi-steel hangers 8 to 9 inches center to center of pins. The gradation screen shall travel with a reciprocating motion imparted by a connecting rod with antifriction bearings at wrist pin and head end, connected to a crankshaft or eccentric shaft having a stroke of 2 to 3 inches. Two locations shall be provided for a crossbar, or the supporting member for wrist-pin bearings, so that when the crossbar is in one location, the hangers supporting the gradation screen will swing to one side of a vertical line, and when located in the outer position the hangers will swing to the opposite side of the same vertical line, reversing the flow of material from the original direction. The screen crankshaft or eccentric shaft shall be of molybdenum steel and shall be mounted in housings equipped with antifriction bearings having grease seals to exclude dust and water. The screen crankshaft or eccentric shaft shall rotate at a maximum speed of 260 rpm. The crankshaft main bearing, the crank-pin bearing, and connecting-rod wrist-pin bearing shall be equipped with flexible grease-tube connections to provide for lubrication from stationary pressure fittings while in operation. The screen crankshaft or eccentric shaft shall be driven by a drive as specified in 3.9 from the roll-crusher countershaft or from any other drive shaft on the unit. The screen crank-shaft or eccentric shaft may be used, if necessary, to drive other mechanisms on the unit.

3.25.6.4 Roll crusher. The roll crusher shall be constructed as specified in 3.25.4.4, except as follows. The rolls shall be not smaller than 40 inches in diameter with not less than 22-inch smooth faces; the shafts shall be not

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less than 7 inches in diameter at the bearings; the slide roll shaft shall be adjustable so that the space between rolls may be varied over a total range of not less than 3-1/2 inches including a negative adjustment of 1 inch; the roll shells shall be not less than 2 inches in thickness; the inside and outside diameters of the shell shall be concentric within 1/16 inch; and the maximum opening between shells, when mounted, shall be not over 1/8 inch when both shells touch at one point during one revolution.

3.25.6.5 Power unit for unit No. 4. The power unit for supplying the power shall be mounted on the rear over the conveyor and shall be supported by means of a heavy frame attached to the main chassis of the crusher unit. The power unit shall be mounted on sliding plates and shall have longitudinal screws for adjusting the belt tension. The fuel tank shall be secured to the chassis and connected to the engine by means of copper tubing. The power shall be transmitted by a V-belt drive as specified in 3.9.2 to the main drive shaft of the crusher.

3.25.6.6 Settling tank. The settling tank shall be built into the semi-trailer frame of unit No. 4 and shall be not less than 36 inches wide by 15 feet long, with sloping ends to reduce the bottom opening to the dimensions required for bolting onto the settling tank of the screw dehydrator unit No. 8. The tank shall be placed in position to receive the sand, silt, mud, and water from the horizontal gradation screen. A longitudinal deflector the full length of the tank shall be bolted to the back sheet of the tank at a point above the spillway. This deflector shall extend down not less than 24 inches into the water and into the center of the tank for the purpose of directing the fine sand down to the bottom of the baffle in order to permit the retention of fine sand. The amount of retention shall be controlled by increasing or decreasing the volume of water.

3.25.6.7 Crushed-rock delivery conveyor. A crushed-rock delivery conveyor shall be mounted below the roll crusher so as to receive the materials passing between the crushing rolls and deliver them to the spout of a detachable return conveyor for return to the revolving scrubber drum in washing operations, or to the horizontal gradation screen in dry operation. The conveyor shall be provided with a channel-type steel frame. The conveyor shall be driven by a drive as specified in 3.9 from an electric motor rated at not less than 5 horsepower and shall be provided with a reduction drive. The motor control switch shall be mounted on the frame of unit No. 4 and shall be accessible for operation from the ground level. Belt carriers shall be as specified in 3.10.2.3.

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3.25.6.8 Delivery hopper. A delivery hopper with supporting legs shall be provided and shall be constructed so that the materials received from the crushed-rock delivery conveyor will drop into the hopper and be directed by by means of a spout into the receiving hopper on the detachable return conveyor. The hopper shall be mounted on structural steel legs that will support the hopper under all operating and transit conditions as specified herein without damage or permanent deformation.

3.25.6.9 Detachable return conveyor. The detachable return conveyor shall be arranged to receive materials from the longitudinal crushed-rock delivery conveyor and the delivery hopper and deliver them either to the feeding spout on the revolving screen when operating as a washing plant, or to the horizontal gradation screen when that screen is operating in reverse direction and the revolving scalping screen is detached or not in use, or to the top deck of the vibrator screen of the roll crusher (unit No. 6). The detachable return conveyor shall be powered by means of an electric motor rated at not less than 10 horsepower through a drive as specified in 3.9 and shall be provided with a reduction drive. The motor-control switch shall be mounted on the frame of the detachable return conveyor and shall be accessible for operation from the ground level. A tripod for supporting the conveyor at the lower end and a supporting rod at the upper end for connection to unit No. 4 shall be provided.

3.25.6.9.1 Conveyor. The conveyor shall be not less than 50 feet long, frame end to frame end, shall be knock-down truss frame or equivalent construction, and shall be divided into three or more sections to facilitate handling in transit. If a vehicle chassis is provided, provision shall be made to remove the chassis when the conveyor is set up in place. Belt carriers shall be as specified in 3.10.2.3.

3.25.7 Unit No. 5, control panel. A power control panel, mounted on a suitable framework, that will receive power from a separate source and distribute it to each of the motors shall be furnished. The panel shall consist of a main contactor to apply and to remove power from the distribution circuits; a power distribution panel with a power connection for each motor; and suitable receptacle connectors for distributing power from the distribution panel to the several motors, and for permitting control of the main contactor from each of the motor starting stations as specified in 3.8.2.1. The control panel may consist either of two units, one for the main contactor and the second for the power distribution panel; or may be a single unit to house both the contactor and the distribution panel.

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3.25.7.1 Main contactor. The main contactor shall be a circuit breaker conforming to NEMA Standard AB1. The circuit breaker shall be a nonautomatic circuit interrupter with shunt trip device. It shall be watertight type IV and dusttight type V in accordance with NEMA Standard IC1. The breaker shall be 3-pole, rated for 225 amperes, and actuated by an external handle. The switch shall be labeled "OFF," "ON," and "RESET."

3.25.7.2 Distribution panel and branch circuit protective devices. The panel shall be watertight type IV and dusttight type V in accordance with NEMA Standard IC1. The panel shall be 3-phase, 4-wire, solid neutral with 300-ampere main-lug capacity. Each branch circuit shall be provided with a circuit breaker in accordance with NEMA Standard AB1. Single-pole circuit breakers will be acceptable if provided with handle tie bars for each individual polyphase circuit. Each starter shall be considered a branch circuit, and individual circuit breakers shall be provided for all starters.

3.25.7.3 Receptacle connectors. Weatherproof receptacle connectors shall be provided on the control panel as follows:

(a) One main power supply receptacle, to mate with the plug connector specified in 3.8.2.2.1. The receptacle shall have bolts attached with wing nuts for clamping on the cover, to provide a weatherproof unit. The cover shall be attached to the aluminum housing by a chain. The receptacle housing shall be connected to the ground conductor and the ground pin.
(b) Motor supply receptacles, as required, to mate with the plug connectors specified in 3.8.2.3. Covers shall be spring-loaded.
(c) Main contactor control receptacles, as required, to permit control of the main contactor from the master OFF pushbutton stations specified in 3.8.2.1. Receptacles shall be 3-pole female and shall be provided with spring-loaded covers.

3.25.8 Unit No. 6, crusher, roll-type, intermediate. The intermediate roll-type crusher unit, for making additional fines in a quarry setup, shall consist of a chassis upon which are mounted a double-deck, vibrator screen; a roll-crusher not less than 40 inches in diameter with a face not less than 22 inches; a power unit; a crushed-rock delivery conveyor; a side-feeder conveyor for pea rock; a side-delivery conveyor for fines; all the necessary combinations of drives, guards, ladders, catwalks, spouts, and hoppers; and all necessary attachments. The unit shall be constructed so that the materials which have passed through the roll crusher on unit No. 4 will be delivered by the detachable return conveyor specified in 3.25.6.9 to the top deck of the vibrator screen. The materials that pass through both the top and bottom decks of the vibrator screen shall be delivered by means of the fines-delivery

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conveyor into the settling chamber of the twin-screw-feed sand dehydrator (unit No. 8) attached to unit No. 4. The materials passing over the top and bottom decks of the vibrator screen shall be delivered to the hopper over the roll crusher (unit No. 6). After passing through the roll crusher (unit No. 6), the crushed material shall be delivered by means of the crushed-rock delivery conveyor under the rolls to the belt conveyor (unit No. 7) specified in 3.25.9 for return by means of hopper and spout to the transfer belt conveyor (unit No. 3), and then to unit No. 4 and unit No. 6, closing the circuit.

3.25.8.1 Chassis. The chassis shall be of the semitrailer type (see 3.11 or 3.12).

3.25.8.2 Vibrator screen. The vibrator screen shall be 4 feet wide and 10 feet long, with two decks of screens, having 3/4-inch square openings on the top deck and 1/4-inch square openings on the bottom deck, and shall be as specified in 3.25.4.3, except that the electric motor shall be rated at not less than 7-1/2 horsepower. Wire screen shall be as specified in 3.4.3. The vibrator screen shall be furnished with spray bars to keep screens clean at all times. Valves to control water at the spray bar shall be furnished, and provision shall be made for connecting the water line into the main water feed of unit No. 4. The wash water shall be discharged into the settling chamber of unit No. 8.

3.25.8.3 Roll crusher. The roll crusher shall be as specified in 3.25.4.4, except for the following changes: The rolls shall be not smaller than 40 inches in diameter with 22-inch smooth faces; the shafts shall be not less than 7 inches in diameter at the bearings; sliding roll space adjustment of not less than 3-1/2 inches including a negative adjustment of 1 inch; the roll shells shall be not less than 2 inches thick; and the inside and outside diameters shall be concentric within 1/32 inch so that the maximum opening between shells, when mounted, will not be over 1/16 inch when both shells touch at one point on the face during one revolution.

3.25.8.4 Power unit for unit No. 6. The power unit for driving the unit shall be mounted above the crushed-rock-delivery conveyor and shall be supported by means of a frame attached to the main chassis of the unit. The power from the engine shall be transmitted by a V-belt drive as specified in 3.9.2 to the main drive shaft of the roll crusher and then to the conveyor countershaft by a drive as specified in 3.9.

3.25.8.5 Crushed-rock-delivery conveyor. A crushed-rock-delivery conveyor shall be mounted longitudinally on the main chassis below the 40-inch-diameter by 22-inch-face roll crusher so as to receive the materials spilled from the crushing rolls and deliver them forward beyond the end of the main chassis

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to the hopper on the transfer belt conveyor (unit No. 7) for return to unit No. 4 by the belt conveyor (unit No. 3). The conveyor shall be provided with a channel-type steel frame. Unless the conveyor has sufficient ground clearance when traveling (see 3.1.2), a raising device shall be provided to raise or fold the conveyor into transit position and return it into operating position. The conveyor shall be driven by a drive as specified in 3.9 from the roll-crusher countershaft through a reversing-gear countershaft. Belt carriers shall be as specified in 3.10.2.3.

3.25.8.6 Side-feeder conveyor. The side-feeder conveyor for pea rock shall be mounted on one side at right angles to the chassis, with its receiving hopper arranged to receive pea rock from the gradation-screen delivery spout on unit No. 4 and shall deliver the materials into the hopper over the roll crusher on unit No. 6. Unit No. 4 and unit No. 6 shall be set parallel and adjacent to each other during operation. The conveyor shall have a channel-type steel frame of folded construction. The conveyor shall be hinged and provided with a raising device for folding into transit position and returning into operating position. The conveyor shall be driven by a sprocket-chain and bevel-gear drive from the roll-crusher countershaft, or from any equivalent drive on the unit. Belt carriers shall be as specified in 3.10.2.3.

3.25.8.7 Side-delivery conveyor. The side-delivery conveyor for fines shall be mounted on one side and at right angles to the chassis, shall be arranged to receive fine materials that have passed through both decks of the vibrator screen on unit No. 6, and shall deliver the fine materials into the settling chamber below the gradation screen on unit No. 4 for delivery to the dehydrator (unit No. 8). Unit No. 4 and unit No. 6 shall be set parallel and adjacent to each other during operation. The conveyor shall have a channel-type steel frame of folded construction. The conveyor shall be hinged and provided with a raising device for folding into transit position and returning into operating position. The conveyor shall be driven by a drive as specified in 3.9 from the roll-crusher countershaft, or from any equivalent drive on the unit. Belt carriers shall be as specified in 3.10.2.3.

3.25.9 Unit No. 7, conveyor, belt. The 30-inch by 35-foot belt conveyor shall be arranged to receive materials from the crushed rock delivery conveyor on the roll crusher (unit No. 6), deliver them to a self-supporting delivery hopper with spout, and then to the hopper on the lower end of the transfer belt conveyor (unit No. 3). The conveyor shall consist of a hydraulic cradle chassis equipped with supporting and hydraulic mechanism upon which is mounted a 30-inch by 35-foot long conveyor, an electric motor rated at not less than 7-1/2 horsepower, a drive as specified in 3.9, a back stop to

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prevent roll back of conveyor belt, and a receiving hopper. The motor control switch shall be mounted on the frame of the unit and shall be accessible for operation from the ground. The belt carriers shall be as specified in 3.10.2.1.

3.25.9.1 Delivery hopper. A delivery hopper, with spout, shall be provided to receive the materials from the conveyor and deliver them to the belt conveyor (unit No. 3). The hopper and spout shall be constructed of sheet or plate steel. The hopper and spout shall be mounted on structural-steel legs of sufficient strength to support them under all operating and transit conditions as specified herein.

3.25.9.2 Chassis. The hydraulic cradle chassis shall be as specified in 3.25.5.1, except that the height of structural-steel supporting members shall permit raising the 35-foot conveyor to the top of the delivery hopper.

3.25.10 Unit No. 8, dehydrator, sand, twin-screw. The twin-screw sand-dehydrator unit for dewatering of sand from unit No. 4 shall be mounted on a separate demountable semitrailer or four-wheel full trailer chassis and shall be capable of attaching to unit No. 4. The unit shall have a chamber not smaller than 48 inches wide, 24 inches deep, and 22 feet long, set at an angle not greater than 20 degrees from the horizontal and equipped with twin screws not less than 22 inches in diameter and 20 feet long. The tank of the screw dehydrator shall be equipped with a weir on the side of the tank near the main truck frame of unit No. 4, shall be located at the same level as that of the spillway in the tank of unit No. 4, and shall be equipped with an adjusting plate for raising, lowering, and leveling the weir. The waste water passing over this weir shall be directed to the waste water flume. When operating, the sand-dehydrator unit shall be located at right angles to the main chassis of unit No. 4. The twin screws, one right-hand and one left-hand, revolving in opposite directions, shall be equipped with removable chilled-iron spiral sections mounted on pipes with shaft ends. The underwater ends of the spiral shafts shall be mounted in rubber or fiber bearings lubricated with fresh water at not less than 10 psi pressure. All other bearings shall be antifriction bearings. The unit shall be driven by a drive as specified in 3.9 from an electric gear-head motor, not less than 20 hp, 440-volt, 60-cycle, 3-phase.

3.25.10.1 Chassis. The chassis shall be of the semitrailer type or a full trailer with a dolly having a fixed type lower fifth wheel. With the exception of the upper and the lower fifth wheels, the chassis and vehicular components of unit 8 shall be as specified in 3.11 or 3.12.

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3.25.11 Unit No. 9, feeder, reciprocating-plate. The reciprocating-plate feeder unit, when delivering materials to unit No. 2, shall consist of a reciprocating-plate feeder with loading hopper and trap gate attached to a detachable 30-inch by 21-foot belt conveyor. When delivering materials to the transfer belt conveyor (unit No. 3), the belt conveyor of unit No. 9 may be omitted and the reciprocating-plate feeder with loading hopper and trap gate shall be mounted over the lower end of unit No. 3. The reciprocating-plate feeder shall be arranged to receive materials through a trap gate or grizzly and deliver the materials at a selected tonnage per hour onto the belt of the detachable belt conveyor. The reciprocating-plate feeder shall consist of a hopper not less than 7 by 7 feet square equipped with a trap gate or grizzly having openings not less than 6 by 8 inches mounted over a reciprocating plate not less than 28 inches wide. The reciprocating plate shall be supported upon not less than two pairs of rollers rotating on shafts mounted in antifriction bearings. The reciprocating plate, rollers, shafts, bearings, and hoppers shall be supported by structural legs and frame and arranged to connect to the detachable belt conveyor so that it can be detached. Reciprocating motion shall be imparted to the plate by means of a crankshaft running in antifriction bearings, transmitting the motion to the plate by means of a connecting rod and wrist pin. The maximum stroke of the reciprocating feeder shall be not less than 6 inches. When the feeder is driven by an eccentric, the feeder shall have not less than four stroke adjustments; and when driven by a countershaft, the feeder shall have not less than four speed adjustments by the use of a combination of sprockets. The drive to the crankshaft or eccentric shaft shall be as specified in 3.9 from the tail pulley of the detachable-belt conveyor, or from a countershaft. The hopper on the reciprocating-plate feeder shall have a spring-retained swinging gate equipped with a quadrant having adjustable stops for limiting the size of the feed opening and so arranged that when larger rocks and foreign articles pass through, the spring tension will permit the gate to swing outward releasing the obstruction, or shall have any equivalent construction of gate provided with an overload device on the gate or drive.

3.25.11.1 Detachable-belt conveyor. The detachable-belt conveyor shall be of channel steel frame not less than 21 feet long. The conveyor shall be driven by a drive as specified in 3.9, powered by an electric motor having a rating of not less than 7-1/2 horsepower, and shall be provided with a suitable reduction drive. The motor control switch shall be accessible for operation from the ground level. The belt conveyor shall be capable of extending the feeder conveyor on unit No. 2, or for transferring material from unit No. 4 to unit No. 10. The belt carriers shall be as specified in 3.10.2.3.

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3.25.12 Unit No. 10, scrubber and washer, gravel, twin-paddle-type. The twin-paddle-type scrubber and washer unit shall receive crushed and washed aggregate from the conveyor specified in 3.25.11.1; subject the aggregate to additional scrubbing, abrading, and washing operations; and deliver the aggregate to the belt conveyor (unit No. 12). The unit shall consist of a steel chassis upon which are mounted a twin-paddle washer having square legs fabricated from steel angles, a power unit, a set of spray pipes, a drive as specified in 3.9, a fuel tank, and all necessary guards, catwalks, drives, valves, and attachments to make the unit complete.

3.25.12.1 Chassis. The chassis of the scrubber and washer shall be of the semitrailer type (see 3.11 or 3.12).

3.25.12.2 Twin-paddle washer. The twin-paddle washer shall consist of a steel tank not less than 7 feet wide and 20 feet long, with sides and bottom sheets not less than 3/8 inch in thickness, set on the chassis at an angle to assure the flow of materials with the chassis in a horizontal position. Within the tank shall be mounted two structural-steel shafts. Each shaft shall support paddles in four rows, 90 degrees apart. The paddles shall be of abrasion- and corrosion-resistant nickel cast iron and shall be replaceable. The shafts shall run in rubber or fiber bearings at the lower ends, shall be lubricated with fresh water under a pressure of not less than 15 psi, and shall be run in antifriction bearings at the upper end. The paddles shall have a radius of not less than 18 inches, shall be not less than 6 inches wide, shall be set at an angle of 60 degrees with the longitudinal axis of the rotating shafts, and shall be reversed on one shaft with respect to those on the other. The shafts shall be not less than 17 feet long from lower paddle to upper paddle, shall rotate in opposite directions, and shall be driven from the power unit by means of a V-belt drive as specified in 3.9.2. The tank shall have a bottom spray pipe not less than 10 feet long located between the two paddle shafts. A main water inlet shall be provided for introducing water into the material from the bottom. A quick-opening drain plug shall be furnished at the lower end of the tank. A control valve shall be furnished for the main water inlet. Hose, nipples, and connections shall be furnished for receiving water from the main waterline on the roll crusher (unit No. 4).

3.25.12.3 Power unit for unit No. 10. The power unit for driving the scrubber and washer unit shall be mounted above the washer tank on the chassis. The power from the engine shall be transmitted to the scrubber and washer by means of a V-belt drive as specified in 3.9.2. The belt drive shall be provided with a spring-tension idler.

3.25.13 Unit No. 11, pump, unit, centrifugal. The centrifugal pumps shall conform to MIL-P-10360, type I, size 6, trailer mounted.

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3.25.14 Unit No. 12, conveyor, belt, coarse material. The 30-inch by 50-foot coarse material belt conveyor shall be arranged to receive crushed rock from the rock delivery hopper on unit No. 4 or unit No. 10, and to discharge the rock to a stock pile. Unit No. 12 shall conform to the requirements specified for unit No. 3 (see 3.25.5) and shall be of the cart-trailer type (see 3.11.9 or 3.12.8).

3.25.15 Unit No. 13, conveyor, belt, sand. The 30-inch by 50-foot sand belt conveyor shall be arranged to receive fines from the delivery spout of the dehydrator (unit No. 8) and deliver the fines to a stock pile when the plant is used under wet operations, or shall be used for receiving fines from a baffle within the settling chamber under the horizontal gradation screen of unit No. 4 when unit No. 6 is removed and unit No. 4 is used as a dry crushing plant. Unit No. 13 shall conform to the requirements specified for unit No. 3 (see 3.25.5) and shall be of the cart-trailer type (see 3.11.9 or 3.12.8).

3.25.16 Unit No. 14, piping equipment. The piping equipment to complete the water-supply system shall consist of a suction hose with terminal pipe, elbow, riser, foot valve, and strainer; supporting ponton for connection to the suction flange on the pump (unit No. 11); a pressure pipe line with valves and fittings; and a water-waste flume.

3.25.16.1 Suction hose. The suction hose shall be 6 inches in diameter, 30 feet long, conforming to ZZ-H-561, grade B, type I, with 6-inch threaded pipe nipples vulcanized into each end and equipped with flanges for bolting to the pump and terminal pipes. A terminal pipe, 6 inches in diameter by 10 feet long equipped with an elbow and a 1-foot-long by 6-inch-diameter riser pipe, shall be fastened to the top platform of the supporting ponton so that the riser pipe may be swung from a vertical position to a horizontal position beyond the edge of the ponton. A foot valve conforming to MIL-V-16721 shall be mounted on the lower end of the riser pipe.

3.25.16.2 Ponton. The ponton for supporting the suction hose line shall consist of a wooden platform not less than 8 by 10 feet supported by four steel drums of not less than 50-gallon volumetric capacity. The steel drums shall be anchored to the ponton by U-shaped bars of not less than 5/8-inch diameter.

3.25.16.3 Pressure pipe line. The pressure pipe line shall consist of a 6-inch spiral riveted or welded pipe having a total length of 2,000 feet, made up of pipe sections not to exceed 30 feet in length, with each section provided with quick-detachable, rubber-gasketed joints. Not less than four

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lengths of pipe shall be provided with 6-inch bolted flanges on one end for connection to the 6-inch pipe flanges on unit No. 4, the suction hose line, or the pump (unit No. 11). The spiral pipe shall be 0.062 inch in nominal wall-thickness and shall be coated inside and outside with asphaltum. Valves and fittings shall be provided in the pipe line where necessary to shut off or regulate the flow of water and complete the water supply system.

3.25.16.4 Waste-water flume. The waste-water flume shall consist of 10 sections of semicircular open-top flume, 10 feet long, not less than 16 inches wide, and made of steel not less than 0.050 inch in nominal thickness, with bolted lap joints.

3.25.17 Unit No. 15, electric generator set. The generator set shall conform to MIL-G-52461; and shall be furnished by the supplier or will be furnished by the Government as specified (see 6.2).

3.26 Workmanship. The plant shall be free from defects such as incomplete welds, rust, cracks, and other defects that could impair the operation or serviceability of the plant.

3.26.1 Steel fabrication. Steel used in the fabrication of the plant shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Corners shall be square and true. Flame cutting may be employed instead of shearing or sawing. Surfaces of flame-cut material, if burned, shall be ground or machined sufficiently to remove ash and cooling checks. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

3.26.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lock-washers shall be provided and all bolts, nuts, and screws shall be tight.

3.26.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven hot with pressure tools and shall completely fill the holes. Rivet heads shall be full, concentric with the rivet holes, and in full contact with the surface of the member.

3.26.4 Welders and welding.

3.26.4.1 Welders. Before assigning any welder to work covered by this specification, the supplier shall provide the contracting officer with

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certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code.

Standard Qualification Procedure of the American Welding Society.
Welding Qualifications of the ASME Boiler and Pressure Vessel Code.

3.26.4.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the plant is tested as specified herein.

3.26.5 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein and on the applicable drawings. Unless prior approval is obtained from the contracting officer, all tapped holes shall be countersunk before tapping.

3.26.5.1 Line boring for main bearings on jaw crusher and roll crushers. Bearing support surfaces for each shaft shall be machined in a single setting to a tolerance of not more than 0.001 inch total indicator reading per foot of shaft. External bearing supports shall be doweled or shall be provided with an eccentric arrangement to assure reassembly to its original position prior to final machining and installation of the bearings.

3.26.6 Surfaces. All parts, components, and assemblies, including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts, shall be cleaned and free of sand, dirt, fins, pits, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth, and all edges shall be rounded or beveled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.1.2 Pump, unit No. 11. Unit No. 11 shall be tested in accordance with MIL-P-10360 and in conjunction with the capacity test specified in 4.6.2.1 through 4.6.2.1.4.

4.1.3 Generator, unit No. 15. Unit No. 15, when furnished by the supplier, shall be tested in accordance with MIL-G-52461 and in conjunction with the capacity tests specified in 4.6.2.1 through 4.6.2.1.4. When furnished by the Government (see 3.25.17) unit No. 15 shall be tested only in conjunction with the capacity tests specified in 4.6.2.1 through 4.6.2.1.4.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The plant shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The plant shall be tested as specified in 4.6.2 through 4.6.2.7. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. Unless otherwise specified (see 3.3), an initial production plant will be selected at random by the Government from the first five plants being produced from production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2.1 and 4.6.2.3 through 4.6.2.6 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of the initial production plants shall not exlude the remaining plants from the quality conformance inspection and acceptance provisions specified in section 4.

4.4.1 Inspection failure. Failure of the initial production plant to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production plant and shall be cause for refusal by the Government to continue acceptance of production plants until evidence has been provided by the

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supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on plants previously accepted and produced under the contract. Any deficiencies found as a result of the initial production testing will be considered prima facie evidence that all plants accepted prior to completion of initial production testing are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each plant shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Test. Each plant shall be tested as specified in 4.6.2.8. Failure of the test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. The plant shall be examined for the following defects. Examination for defects 114 through 118 shall be made while the plant is operating.

101. Material not as specified.
102. Parts or components missing or do not function.
103. Assembly incomplete or incorrect.
104. Safety devices or guards missing.
105. Workmanship not as specified.
106. Ease of maintenance not as specified.
107. Dimensions or weight not as specified.
108. Treatment and painting not as specified.
109. Identification or other marking, operating, and safety instructions, or manuals missing.
110. Lubrication devices not accessible.
111. Foundation brake protection not as specified.
112. Brake system not as specified or does not function.
113. Electrical system not as specified, does not function, or is damaged.
114. Oil or grease seals leak.
115. Bearings or gear boxes overheat.
116. Shaft misalinement.
117. Belt carrier rolls do not turn.
118. Belt does not run straight on the centerline of the conveyors.
119. Nonconformance to ICC regulations.

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4.6.2 Tests. Prior to test, lubricate the plant with Military oils and greases as specified in 3.15. Oils shall be those designated for use in the ambient temperature at which the test is being performed.

4.6.2.1 Crushing capacity.

4.6.2.1.1 Test conditions.

4.6.2.1.1.1 Net operating time. Crushed-rock production capacity shall be computed on the basis of net operating time, which shall be defined as gross testing time less allowable downtime for the following:

(a) Jamming from rock fragments larger than maximum size of rock specified in 3.25.3.
(b) Normal machinery downtime. Downtime due to any design deficiency shall not be considered normal machinery downtime.
(c) Delay due to lack of material delivery.
(d) Other conditions considered by the contracting officer to be beyond the control of the supplier.

4.6.2.1.1.2 Aggregate quantities. For the purpose of determining the quantities of graded aggregates produced during the tests, only that amount of material shall be considered that falls between limiting screen sizes, except for the tolerances specified.

4.6.2.1.1.3 Percentage of crush. Arrange the flow of material so that all material passing through the vibrating grizzly bars on jaw crusher unit No. 1 will be discharged by the side delivery conveyor on jaw crusher unit No. 1. Weigh not less than 300 tons of material to be processed during the capacity tests. The material bypassed through the grizzly bars shall be weighed separately and should not exceed 20 percent of the combined total of material discharged by the side delivery conveyor on the jaw crusher unit No. 1 and the crushed material processed by the plant. Presence in the feed of more than 20 percent of material not requiring crushing for a period of more than 2 consecutive hours shall constitute failure of this test. This test may be conducted in conjunction with 4.6.2.1.2, 4.6.2.1.3, or 4.6.2.1.4.

4.6.2.1.1.4 Standards of accuracy. Test apparatus accuracy shall be as follows:

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Type Of Measurement	Percent Of Measured Value
(a) Deceleration	± 2
(b) Force	± 2
(c) Linear dimension	± 1
(d) Speed	± 2
(e) Weight	± 0.5

4.6.2.1.2 Washed material. Operate the plant to produce washed material for 40 hours processing rock as specified in 3.25.1 that contains soil as specified in 3.25.1.2. Inability to produce an average of 225 tons per hour proportioned as specified in 3.25.1.2, or any malfunction, permanent deformation, or breakage shall constitute failure of this test.

4.6.2.1.3 Dry material. Operate the plant to produce dry material for 10 hours processing rock as specified in 3.25.1. Inability to produce an average of 125 tons per hour proportioned as specified in 3.25.1.3, or any malfunction, permanent deformation, or breakage shall constitute failure of this test.

4.6.2.1.4 Dry material gross capacity. Operate the plant for 10 hours processing rock as specified in 3.25.1. Inability to produce an average of 300 tons per hour proportioned as specified in 3.25.1.4, or any malfunction, permanent deformation, or breakage shall constitute failure of this test.

4.6.2.1.5 Flywheel. Continuously measure the speed of the flywheel on the jaw crusher (unit No. 1) during the capacity tests for a period of 3 minutes. Select a 3-minute period during which at least one large rock of near capacity size is being crushed. Nonconformance to 3.25.3.5.6 shall constitute failure of this test.

4.6.2.1.6 Uncrushable material. With the jaw crusher (unit No. 1) operating at capacity rate, drop a piece of uncrushable material, such as tramp iron, into the crusher. The piece shall be larger than the maximum jaw opening at the jaw setting being used, but not more than 2 inches larger. Nonconformance to 3.25.3.5.7 shall constitute failure of this test.

4.6.2.2 Engine performance. Prior to examination and tests of the plant, the engines that will be installed in the preproduction model units shall be tested in accordance with test method series 2000 of MIL-STD-. Nonconformance to 3.8.1.1 of this specification or failure to meet the requirements of test method series 2000 of MIL-STD-1400 shall constitute failure of this test.

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4.6.2.3 Highway and cross-country. Before conducting any highway and cross-country tests, measure all critical mounting points on each unit. Tow each trailer-mounted unit and transport each nontrailer-mounted unit as follows:

(a) 250 miles over highways at speeds up to 30 mph.
(b) 10 miles cross-country at speeds up to 10 mph.
(c) Demonstrate conformance to 3.7 through 3.7.1.5.

Upon completion of the travel, remove at least one wheel and drum on each unit and examine the entire plant, including the brakes. Check the measurement of critical mounting points and magnaflux all welds. Nonconformance to 3.7 through 3.7.1.5, any change in critical mounting point dimensions, evidence of any weld being defective, any breakage, permanent deformation, or malfunctioning shall constitute failure of this test.

4.6.2.4 Rail transport. Sectionalize the plant for rail transport and reassemble. Nonconformance to 3.7.2 shall constitute failure of this test.

4.6.2.5 Lifting attachments. Lift each unit and section of the plant and determine the normal load on each attachment. Measure the clearance between the slings and the exterior parts of each section and measure the convergence of the slings from the lowest extremity of the load. Apply 2-1/2 times the normal load to each attachment in the direction in which a sling normally applies the lifting load. Nonconformance to 3.16, weld failure, or permanent deformation of the attachment or any part of the plant shall constitute failure of this test.

4.6.2.6 Tiedown attachments. Anchor each section of the plant one at a time to a flatcar using the tiedown attachments and no blocking, bracing, or cribbing. Hump the loaded flatcar at a speed of 7 mph against five boxcars with their airbrakes set. Repeat the humping with the opposite end of the section in the direction of the five cars. Repeat the procedure on each assembled unit. Weld failure or permanent deformation of the attachments or any part of the plant shall constitute failure of this test.

4.6.2.7 Electromagnetic compatibility. The preproduction model, equipped for the reduction of radio interference in accordance with 3.14, shall be tested as specified in MIL-E-55301. The supplier shall furnish the contracting officer the reports required by MIL-E-55301, except that the time interval for submission of reports shall be as specified (see 6.2). Upon approval of the reports by the contracting officer and provided all other requirements of the specification are met, the preproduction model shall be used as a model for all production plants.

4.6.2.8 Operation. Operate the plant for a period of 1 hour without processing material. Any operational difficulties, breakage, or permanent deformation shall constitute failure of this test.

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4.7 Inspection comparison. The Government may select plants at any time during the contract production period and subject these plants to the examination specified in 4.6.1 and the tests specified in 4.6.2.1 and 4.6.2.3 through 4.6.2.6 to determine that the quality of the selected plant is equal to the quality standards established during initial production inspection. The inspection will be performed by the Government at a site selected by the Government. Plants will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production plant.

4.7.1 Inspection failure. Failure of any plant when inspected as specified in 4.7 may be considered cause for refusal by the Government to continue acceptance of plants until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.8 Inspection of preparation for delivery.

4.8.1 Pilot pack inspection.

4.8.1.1 Examination. The pilot pack shall be examined in accordance with 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.1.2 Test. The crated components shall be tested in accordance with 4.8.3. Failure of the railroad-car impact test shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completely processed plant including related crated components shall be considered a unit of product.

4.8.2.2 Sampling. Samples for examination shall be selected in accordance with MIL-STD-105. AQL shall be 2.5 percent defective.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects:

120.	Materials, methods, or containers not as specified. Each incorrect material, method, or container shall be considered one defect.
121.	Sectionalization and disassembly not as specified for level A.
122.	Matchmarking not as specified for level A.

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123. Unpainted exterior metal surfaces not coated with preservative as specified for level A.
124. Preserved machined surfaces not wrapped or covered with barrier material as specified for level A.
125. Power units not preserved in accordance with the referenced documents for level A.
126. Trailer chassis and dolly components not preserved in accordance with the referenced document for level A.
127. Pumps not preserved in accordance with the referenced document for level A.
128. Generator not preserved in accordance with the referenced document for level A.
129. Loose piping equipment components not preserved and packaged in accordance with the referenced documents for level A.
130. Clutches not preserved as specified for the applicable type for level A.
131. Gear cases not filled to operating level as specified for level A.
132. Drive chains not preserved as specified for level A.
133. Drive belts not removed or released from tension for level A.
134. Metal belt lacings not coated with varnish as specified for level A.
135. Conveyor belts not removed or released from tension for level A.
136. Openings into electrical components not sealed for level A.
137. Instrument glasses that are not protected by a housing or cover are not covered with plywood as specified for level A.
138. Power cables not wound on the reels and secured for level A.
139. Hydraulic systems not preserved as specified for level A.
140. Repair parts not preserved and packaged in accordance with referenced document for level A.
141. Maintenance tools not preserved and packaged as specified for level A.
142. Technical publications not preserved as specified for level A.
143. Components requiring protection and not specifically identified not preserved and packaged as specified for similar components, for level A.
144. Intermediate packaging not as specified for level A.
145. Packaged and unpackaged components secured to the units, interfere with lifting or towing the unit for level A.
146. Blocking, bracing, and anchoring not in accordance with the referenced document for level A.
147. Marking illegible, incorrect, or incomplete.

4.8.3 Test. The following test shall be performed on the crated components for level A. At the conclusion of the pilot pack test, the pack shall

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be examined. Any shifting of contents, loose or broken holddowns, ties, stays, blocking, or bracing, or any visible damage to the contents shall constitute failure of the pilot pack.

4.8.3.1 Railroad-car impact test. The crated components shall be subjected to the railroad-car test as specified for guided-impact tests in Appendix A to MIL-STD-1186. The car shall strike a string of five empty cars with draft gear extended and the brakes set, at a speed of between 10 and 11 miles per hour.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. The pilot pack is defined as a preproduction pack consisting of one complete plant, including all related crated equipment, which will be identical to the pack which the supplier will subsequently produce in fulfillment of the contract. Before a number of plants is packed, a pilot pack shall be made, examined, and related crated equipment shall be subjected to the test specified for the applicable level to determine conformance to the preservation, packaging, packing, and marking requirements. Approval of the pilot pack shall not relieve the supplier of his obligation to preserve, package, pack, and mark the plant in accordance with this specification. Any changes or deviations of production packs from the pilot pack shall be subject to the approval of the contracting officer.

5.2 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Sectionalization and disassembly. The plant shall be sectionalized to meet the requirements specified in 3.7.2. In addition, components of the units and sections vulnerable to damage, pilferage, and loss, and protruding parts that increase cubage and that can be removed and reassembled without special tools shall be removed. Removed bolts, nuts, pins, and washers shall be replaced in one of the mating parts and secured to prevent their loss. Removed keys shall be secured in the keyway of the pulley or flywheel.

5.2.1.2 Matchmarking. To facilitate reassembly, removed components and sections shall be matchmarked. Large sections and components shall be matchmarked identically by stencilled letters or numerals with lusterless white enamel, overcoated with varnish conforming to TT-V-121; or identically matchmarked with paint conforming to MIL-P-13983. Removed small components shall have matchmarking information on cloth tags conforming to UU-T-81, type A, or on metal tags. The cloth tags shall be waterproofed in accordance with MIL-STD-129. The tags shall be securely attached to the components so as to prevent damage to the preservation or to the components.

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5.2.1.3 Preservatives. Preservatives specified herein shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.2.1.4 Unpainted surfaces. Unpainted exterior metal surfaces of components of the plant such as shafts, rollers, belt tension screws, idlers, frame members, flumes, sprockets, spray bars, springs, nonprecision chains, tie rods, surfaces exposed by sectionalization or disassembly, or any other similar unpainted exterior metal surface not definitely specified herein requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.2.1.5 Machined surfaces. Unpainted exterior machined metal surfaces shall be coated with type P-6 preservative. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, grade A, type II, class 2 and the material shall be secured in place with tape conforming to PPP-T-60, class 1.

5.2.1.6 Power units.

5.2.1.6.1 Engines. Diesel engines, gasoline engine (unit No. 11), and engine accessories shall be preserved in accordance with the level A requirements of MIL-E-10062.

5.2.1.6.2 Electric motors. The electric motors shall be preserved in accordance with MIL-E-16298, level A, using the alternate method as specified for assembled machines.

5.2.1.7 Trailer chassis. The components of the trailer chassis shall be preserved and packaged in accordance with the level A (mobile) requirements of MIL-STD-281.

5.2.1.8 Pumps. The centrifugal pumps and pump components shall be preserved and packaged in accordance with MIL-P-10360, level A.

5.2.1.9 Generator. The generator set shall be preserved and packaged in accordance with MIL-G-52461, level A.

5.2.1.10 Piping equipment. The loose piping equipment components shall be preserved and packaged in accordance with the level A requirements of MIL-H-775 or MIL-V-3, as applicable.

5.2.1.11 Clutches.

5.2.1.11.1 Disk type. With the cover plates removed and with the clutch engaged, all interior components of the clutch shall be sprayed with primer conforming to TT-P-664. Coating of nonmetallic surfaces shall be kept to

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a minimum. Clutch control mechanism not enclosed shall be coated with type P-1 preservative. After being sprayed, spring-loaded-type clutches shall be blocked in partially disengaged position to eliminate contact between disk facings and pressure plates. Snap-over-center and toggle-in-type clutches shall be completely disengaged.

5.2.1.11.2 Band and shoe type. Clutch drum facings and control mechanisms enclosed within housings shall be coated with primer conforming to TT-P-664. Control mechanisms not enclosed within housings shall be coated with type P-1 preservative. The clutch shall not be operated after application of preservatives.

5.2.1.11.3 Jaw type. Clutch jaws, shifter yokes, and machined surfaces of shafts shall be coated with type P-4 preservative. Clutch control mechanisms enclosed within housings shall be fogged with type P-10, grade 2 preservative. Clutch control mechanisms not enclosed within housings shall be coated with type P-1 preservative. The clutch shall not be operated after application of preservatives.

5.2.1.11.4 Cone type. The unlined cone and clutch control mechanisms shall be coated with a thin film of primer conforming to TT-P-664. The clutch shall be disengaged to prevent the lining from adhering to the unlined cones. The clutch shall not be operated after application of preservatives.

5.2.1.12 Gears.

5.2.1.12.1 Enclosed gears. When the gear lubricant contained in the gear housing conforms to the applicable lubrication order or to MIL-L-2105, or MIL-L-21260, the quantity required to bring the lubricant to the operating level shall be added. When the lubricant does not conform to any of the lubricants specified herein, the gear housing shall be drained and refilled to the operating level with the applicable lubricant required for operation. The gears shall be actuated to assure that all interior parts and surfaces are coated.

5.2.1.12.2 Exposed gears. Exposed gears shall be coated with type P-1 preservative or with primer conforming to TT-P-664.

5.2.1.13 Drive chains.

5.2.1.13.1 Exposed chains. Exposed drive chains shall be coated with enough type P-3 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. The excess preservative shall be drained, and the entire chain shall be coated with type P-1 preservative. Removed chains shall be coiled, and the coil shall be secured with

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soft annealed wire and wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2, secured with tape conforming to PPP-T-60, class 1. Sprockets shall be coated with type P-1 preservative.

5.2.1.13.2 Enclosed chains. Enclosed chains and chain housings shall be preserved as specified in 5.2.1.12.1 for enclosed gears and gear housings.

5.2.1.14 Shafts and universal joints. Exposed surfaces of shafts and universal joints shall be coated with type P-1 preservative.

5.2.1.15 Drive belts. Drive belts shall be removed or released from tention. Removed belts shall be preserved in accordance with MIL-P-116, method IC-1. Pulley grooves shall be coated with primer conforming to TT-P-664. Conveyor belts shall be tagged or identified to indicate their reinstallation positions.

5.2.1.16 Conveyor belts. Metal belt lacings shall be coated with primer conforming to TT-P-664. The belts shall be removed or released from tension. When removed, the belts shall be compactly rolled and the roll shall be secured with flat steel strapping conforming to QQ-S-781, type I, class B, size as applicable.

5.2.1.17 Electrical components (except motors).

5.2.1.17.1 Openings. Openings into the control panel, instrument panels, switches, controls, junction boxes, exposed ends of electric wiring, and similar items shall be sealed with tape conforming to PPP-T-60, class 1. The glass of instruments not protected by a housing or cover shall be covered with a fitted piece of 1/4-inch-thick plywood secured in place with tape specified herein.

5.2.1.17.2 Power cables and reels. Unpainted exterior surfaces of the reels shall be coated with type P-1 preservative. The power cables shall be evenly wound on the reels and secured to prevent unwinding.

5.2.1.17.3 Power cable adapter assembly and auxiliary cable. The power cable adapter assembly and auxiliary cable shall be preserved together in accordance with MIL-P-116, method IC-5.

5.2.1.18 Hydraulic systems. The hydraulic fluid supply reservoir shall be filled to the operating level with the hydraulic fluid recommended for operation. The pistons shall be retracted into the cylinders and secured. Any exposed portions of the piston rods shall be coated with type P-6 preservative and wrapped or covered with barrier material conforming to MIL-B-121,

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type I, grade A, class 2, extending the wrap 2 inches on the ram cylinder. The wrap shall be secured in place with tape conforming to PPP-T-60, class 1. The hydraulic control valves shall be secured in the neutral position and preserved as specified for the piston rods. Hose shall not be disconnected unless required to meet the disassembly requirements. Any other exterior metal surface of components of the system requiring the application of a contact preservative shall be coated with type P-1 preservative.

5.2.1.19 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.2.1.20 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with MIL-H-15424, level A.

5.2.1.21 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1.

5.2.1.22 Miscellaneous preservation and packaging. Components not specifically identified herein requiring protection from corrosion and mechanical damage shall be preserved and packaged as specified for components of similar design or construction.

5.2.1.23 Intermediate packaging. Packaged and unpackaged components shall be placed in the toolbox or container provided on the plant for that purpose. All other packaged and unpackaged components that cannot be secured to the plant and that require the additional protection of a container shall be intermediate packaged in close-fitting boxes conforming to PPP-B-621, class 1, style optional or PPP-B-601, domestic type, style optional. The contents shall be blocked, braced, and anchored within the box in accordance with MIL-STD-1186. The boxes shall be closed in accordance with the appendix to the applicable box specification. Strapping shall not be required.

5.2.2 Level C. Preservation and packaging of the components of the plant shall afford protection against deterioration and damage from the supplier to the initial destination. The supplier may use his standard practice providing it fulfills these requirements.

5.3 Packing. Packing shall be level A or C as specified (see 6.2).

5.3.1 Level A. The plant shall be prepared for mobile shipment. When sufficient space is available, packaged and unpackaged components and sections shall be placed and secured on the unit for which intended in any available space that will not interfere with the lifting or mobility of the unit. The components shall be secured to the unit with zinc-coated strapping conforming

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to QQ-S-781 or QQ-S-790. Packaged and unpackaged components and sections of the plant that cannot be secured to one of the units shall be packed in crates conforming to MIL-C-3774, nailed assembly, skid type, or PPP-C-650, style A, type as applicable. Blocking, bracing, and anchoring of the contents shall be in accordance with MIL-STD-1186. Crated components shall be shipped at the same time as the mobile components of the plant.

5.3.2 Level C. The complete plant shall be prepared for shipment in a manner to assure carrier acceptance and safe delivery to destination at lowest rating. The plant shall be prepared for mobile shipment. When sufficient space is available, packaged and unpackaged components and sections shall be placed and secured on the unit for which intended in any available space that will not interfere with the lifting or mobility of the unit. Packaged and unpackaged components and sections of the plant that cannot be secured to one of the units shall be packed in crates complying with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules applicable to the mode of transportation. Crated components shall be shipped at the same time as the mobile units of the plant.

5.4 Marking. Packages, shipping containers, and mobile units shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This plant is intended to be used in the production of crushed rock and sand from gravel pits and rock quarries for use as aggregates in cement concrete, asphaltic concrete, bituminous mixtures, or compacted road-surfacing material.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When initial production plant is not required (see 3.3).
(c) When engines shall be other than as specified in 3.8.1.
(d) When commercial vehicular components are required (see 3.11).
(e) When an emergency airbrake system conforming to the California Vehicle Code is not required for commerical trailers (see 3.11.6.2).
(f) When a dolly converter is required and dolly characteristics required (see 3.11.8).
(g) When Military vehicular components are not required (see 3.12).
(h) When electromagnetic compatibility other than as specified is required (see 3.14).
(i) Technical publications required (see 3.21).
(j) Repair parts and maintenance tools required (see 3.22).

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(k) Whether unit No. 15 (electric generator set) shall be furnished by the supplier or will be furnished by the Government (see 3.25.17).
(l) Time interval for submission of electromagnetic compatibility reports (see 4.6.2.7 and 6.4).
(m) Level of preservation and packaging and level of packing required (see 5.2 and 5.3).

6.3 Lubricants. The contracting officer should furnish a list of Military lubricants applicable to the plant covered by this specification as contained in the Federal Supply Catalogue, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.4 Electromagnetic compatibility. The contracting officer should submit a copy of the report(s) required in 4.6.2.7 to the Commanding General, U. S. Army Electronics Command, ATTN: AMSEL-RD-GF, Fort Monmouth, New Jersey.

6.5 Revision changes. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - MO(ERDL)
Air Force - 84

Preparing activity:

Army - MO(ERDL)

Review activities:

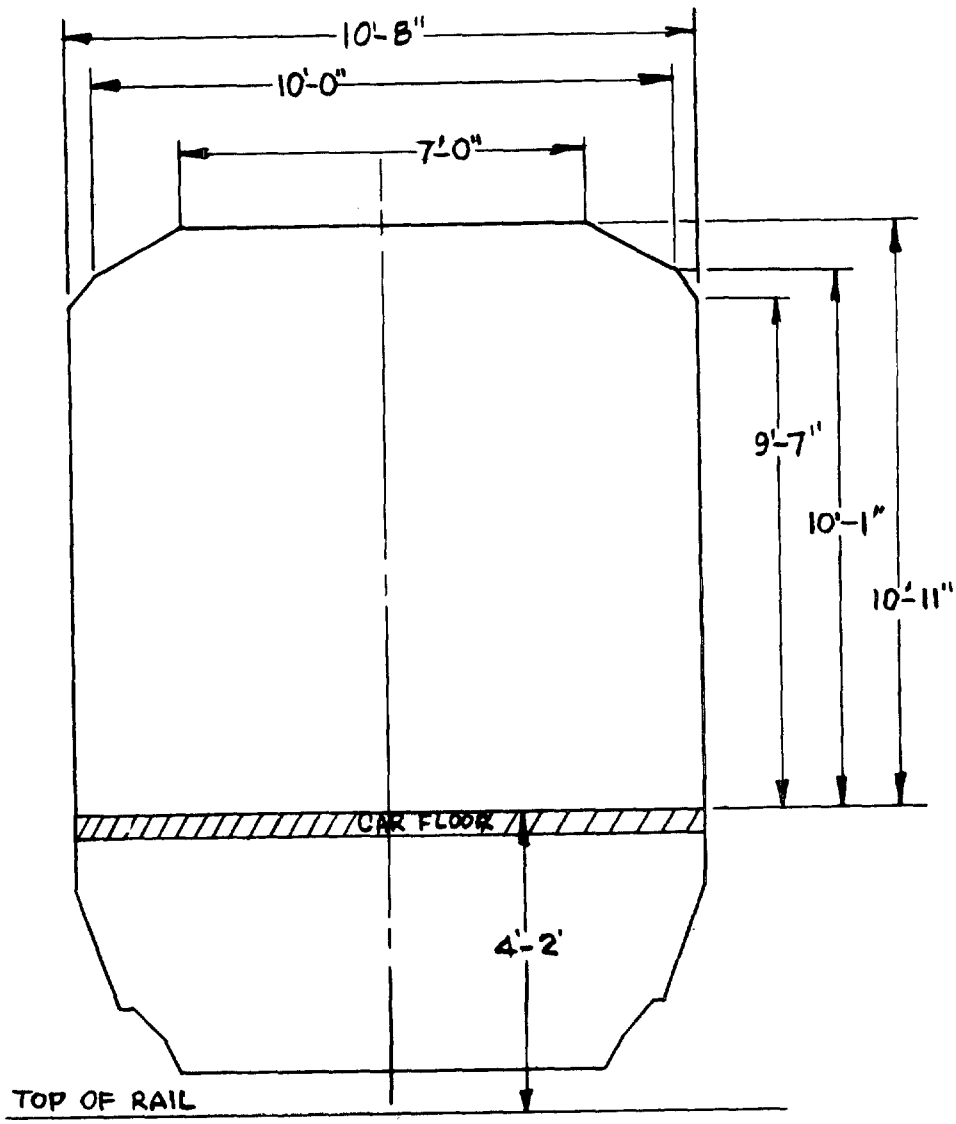
Army - MO, EL
Air Force - 84, 85

User activities:

Army - CE
Air Force - None

Project No. 3820-0051

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NOTE: DIAGRAM COVERS CARS OF DIMENSIONS UP TO:

INSIDE LENGTH	50'-6"
COUPLED LENGTH	54'-8½"
℄ TO ℄ OF TRUCK	41'-3"

Figure 1 OUTLINE DIAGRAM OF APPROVED LIMITED CLEARANCES OF ASSOCIATION OF AMERICAN RAILROADS.

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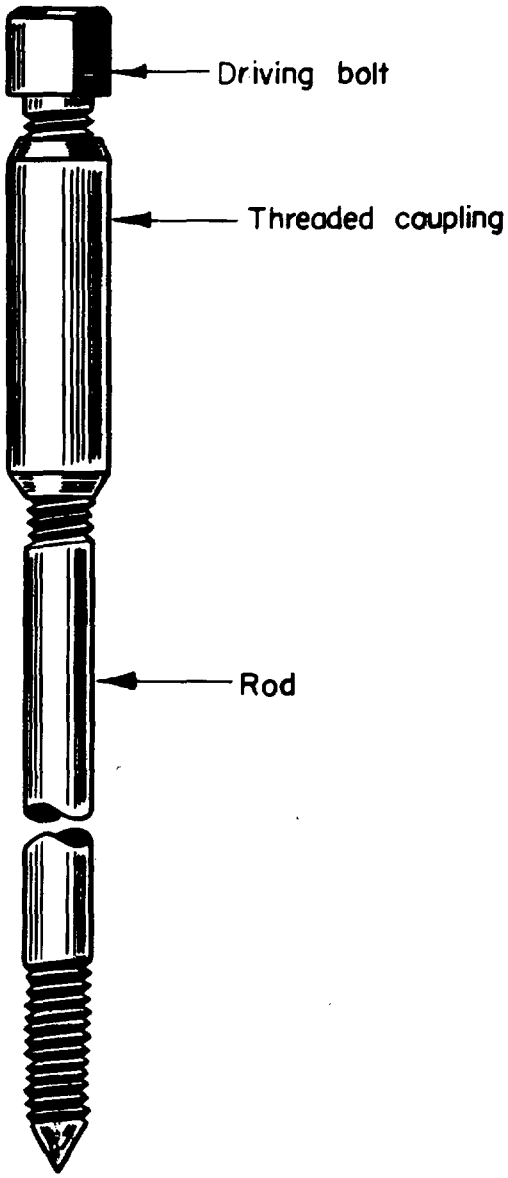
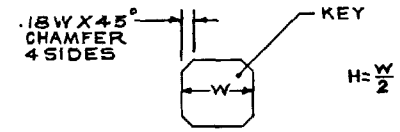
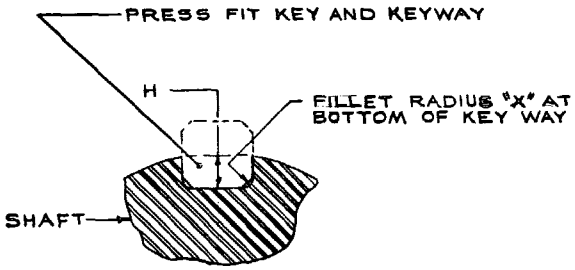


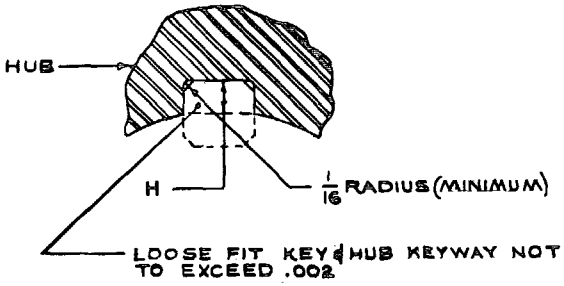
Figure 2 Rod assembly, ground.

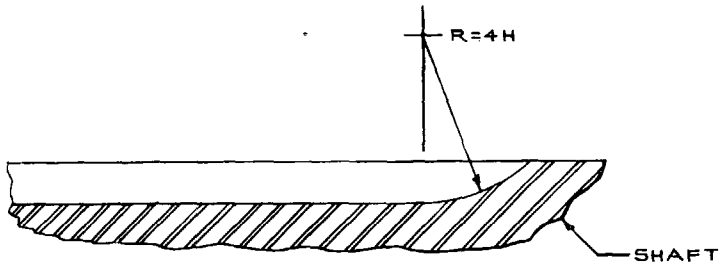
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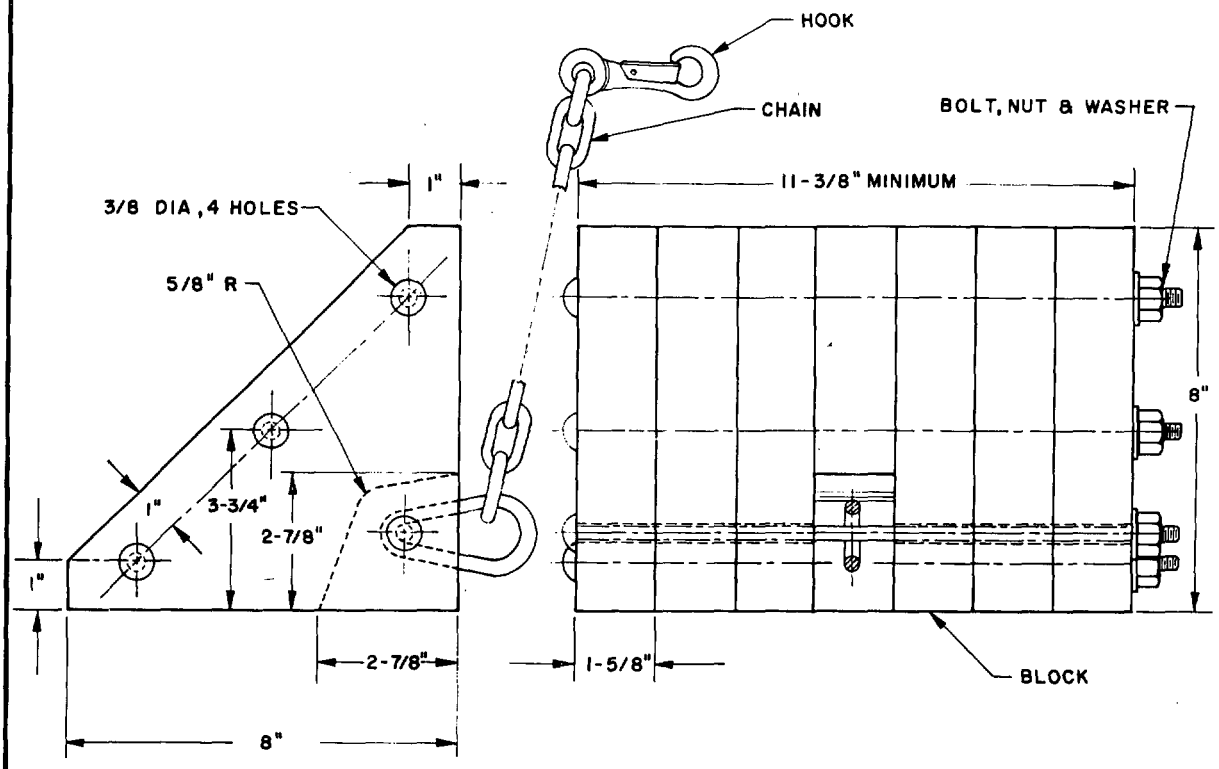
TYPICAL KEYWAY CONTOUR

FILLET RADIUS AT BOTTOM OF KEYWAY	
H	X
UNDER 1/4	SAE
1/4 , 5/16	3/32
3/8	1/8
7/16 , 1/2	5/32
9/16	3/16
5/8 , 11/16	7/32
3/4	1/4
13/16 , 7/8	9/32
15/16	5/16
1 , 1 1/16	11/32
1 1/8 TO 1 1/8	3/8

FIGURE 3 KEYS AND KEYWAYS

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CHAIN DETAILS

CHAIN WELDED, CLOSE STRAIGHT LINK, GALV'D, 1/4" NOMINAL SIZE OF CHAIN BARS.

LENGTH OF 100 LINKS = 100 INCHES

MINIMUM TENSILE STRENGTH = 3,400 LBS.

WEIGHT/100 FT. = 70 LBS.

MATERIAL NOTE

WOOD PARTS, FABRICATED, HARDWOOD.

FIGURE 4 CHOCK BLOCK ASSEMBLY

DX-603

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